2020 ESC Guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS)

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ESC entities having participated in the development of this document:

Associations: Association for Acute CardioVascular Care (ACVC), Association of Cardiovascular Nursing & Allied Professions (ACNAP), European Association of Cardiovascular Imaging (EACVI), European Association of Preventive Cardiology (EAPC), European Association of Percutaneous Cardiovascular Interventions (EAPCI), European Heart Rhythm Association (EHRA), Heart Failure Association (HFA).

Councils: Council on Stroke, Council on Valvular Heart Disease.

Working Groups: Cardiac Cellular Electrophysiology, Cardiovascular Pharmacotherapy, Cardiovascular Surgery, e-Cardiology, Thrombosis.

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ESC Classes of recommendations

	Definition	Wording to use
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

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ESC Levels of evidence

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

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What is new in the 2020 Guidelines? New recommendations (1)

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Recommendations	Class
Recommendations for diagnosis of AF	
ECG documentation is required to establish the diagnosis of AF. A standard 12-lead ECG recording or a single-lead ECG tracing of ≥30 seconds showing heart rhythm with no discernible repeating P waves and irregular RR intervals (when atrioventricular conduction is not impaired) is diagnostic of clinical AF.	I
Recommendations for structured characterization of AF	
Structured characterization of AF, which includes clinical assessment of stroke risk, symptom status, burden of AF, and evaluation of substrate, should be considered in all AF patients, to streamline the assessment of AF patients at different healthcare levels, inform treatment decision-making, and facilitate optimal management of AF patients.	IIa

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What is new in the 2020 Guidelines? New recommendations (2)

Recommendations	Class
Recommendations for screening to detect AF	
When screening for AF it is recommended that: • The individuals undergoing screening are informed about the significance and treatment implications of detecting AF. • A structured referral platform is organized for screen-positive cases for further physician-led clinical evaluation to confirm the diagnosis of AF and provide optimal management of patients with confirmed AF. • Definite diagnosis of AF in screen-positive cases is established only after the physician reviews the single-lead ECG recording of ≥30 seconds or 12-lead ECG and confirms that it shows AF.	I

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What is new in the 2020 Guidelines? New recommendations (3)

Recommendations	Class
Recommendations about integrated AF management	
It is recommended to routinely collect PROs to measure treatment success and improve patient care.	I
Recommendations for the prevention of thromboembolic events in AF	
For a formal risk-score–based assessment of bleeding risk, the HAS-BLED score should be considered to help address modifiable bleeding risk factors, and to identify patients at high risk of bleeding (HAS-BLED score ≥3) for early and more frequent clinical review and follow-up.	IIa
Stroke and bleeding risk reassessment at periodic intervals is recommended to inform treatment decisions (e.g. initiation of OAC in patients no longer at low risk of stroke) and address potentially modifiable bleeding risk factors.	I

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What is new in the 2020 Guidelines? New recommendations (4)

Recommendations	Class
Recommendations for the prevention of thromboembolic events in AF (continued)	
In patients with AF initially at low risk of stroke, first reassessment of stroke risk should be made 4–6 months after the index evaluation.	IIa
Estimated bleeding risk, in the absence of absolute contraindications to OAC, should not in itself guide treatment decisions to use OAC for stroke prevention.	III
Clinical pattern of AF (i.e. first detected, paroxysmal, persistent, long-standing persistent, permanent) should not condition the indication to thromboprophylaxis.	III

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What is new in the 2020 Guidelines? New recommendations (5)

Recommendations	Class
Recommendations for cardioversion	
Pharmacological cardioversion of AF is indicated only in a haemodynamically stable patient, after consideration of the thromboembolic risk.	I
For patients with sick-sinus syndrome, atrioventricular conduction disturbances or prolonged QTc (>500 ms), pharmacological cardioversion should not be attempted unless risks for proarrhythmia and bradycardia have been considered.	III

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What is new in the 2020 Guidelines? New recommendations (6)

Recommendations	Class
Recommendations for rhythm control/catheter ablation of AF	
General recommendations	
For the decision on AF catheter ablation, it is recommended to take into consideration the procedural risks and the major risk factors for AF recurrence following the procedure and discuss them with the patient.	I
Repeated PVI procedures should be considered in patients with AF recurrence provided the patient's symptoms were improved after the initial PVI.	IIa
AF catheter ablation after antiarrhythmic drug therapy failure	
AF catheter ablation for PVI should be considered for rhythm control after one failed or intolerant to beta-blocker treatment to improve symptoms of AF recurrences in patients with paroxysmal and persistent AF.	IIa

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What is new in the 2020 Guidelines? New recommendations (7)

Recommendations	Class
Recommendations for rhythm control/catheter ablation of AF (continued)	
First-line therapy	
AF catheter ablation for PVI should/may be considered as first-line rhythm control therapy to improve symptoms in selected patients with symptomatic: • Paroxysmal AF episodes, or	**IIa**
• Persistent AF without major risk factors for AF recurrence as an alternative to AAD class I or III, considering patient choice, benefit, and risk.	**IIb**
Techniques and technologies	
Use of additional ablation lesions beyond PVI (low voltage areas, lines, fragmented activity, ectopic foci, rotors, and others) may be considered but is not well established.	**IIb**

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What is new in the 2020 Guidelines? New recommendations (8)

Recommendations	Class
Recommendations for rhythm control/catheter ablation of AF (continued)	
Lifestyle modification and other strategies to improve outcomes of ablation	
Strict control of risk factors and avoidance of triggers are recommended as part of rhythm control strategy.	I
Recommendations for stroke risk management peri cardioversion	
It is recommended that the importance of adherence and persistence to NOAC treatment both before and after cardioversion is strongly emphasized to patients.	I
In patients with AF duration of >24 hours undergoing cardioversion, therapeutic anticoagulation should be continued for at least 4 weeks even after successful cardioversion to sinus rhythm (beyond 4 weeks, the decision about long-term OAC treatment is determined by the presence of stroke risk factors).	IIa

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What is new in the 2020 Guidelines? New recommendations (9)

Recommendations	Class
Recommendations for stroke risk management peri cardioversion (continued)	
In patients with a definite duration of AF ≤24 hours and a very low stroke risk (CHA_2DS_2-VASc of 0 in men or 1 in women) post-cardioversion anticoagulation for 4 weeks may be omitted.	**IIb**
Recommendations for stroke risk management peri catheter ablation	
In AF patients with stroke risk factors not taking OAC before ablation, it is recommended that preprocedural management of stroke risk includes initiation of anticoagulation and: • Preferably, therapeutic OAC for at least 3 weeks before ablation, or	**I**
• Alternatively, the use of TOE to exclude LA thrombus before ablation.	**IIa**
For patients undergoing AF catheter ablation who have been therapeutically anticoagulated with warfarin, dabigatran, rivaroxaban, apixaban, or edoxaban, performance of the ablation procedure without OAC interruption is recommended.	**I**

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What is new in the 2020 Guidelines? New recommendations (10)

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Recommendations	Class
Recommendations for long-term AADs	
In AF patients treated with sotalol, close monitoring of QT interval, serum potassium levels, CrCl, and other proarrhythmia risk factors is recommended.	I
In AF patients treated with flecainide for long-term rhythm control, concomitant use of an atrioventricular nodal-blocking drug (if tolerated) should be considered.	IIa
Sotalol may be considered for long-term rhythm control in patients with normal LV function or with ischaemic heart disease if close monitoring of QT interval, serum potassium levels, CrCl, and other proarrhythmia risk factors is provided.	IIb

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What is new in the 2020 Guidelines? New recommendations (11)

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Recommendations	Class
Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in AF	
Identification and management of risk factors and concomitant diseases is recommended as an integral part of treatment in AF patients.	I
Modification of unhealthy lifestyle and targeted therapy of intercurrent conditions is recommended to reduce AF burden and symptom severity.	I
Opportunistic screening for AF is recommended in hypertensive patients.	I
Opportunistic screening for AF should be considered in patients with OSA.	IIa

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What is new in the 2020 Guidelines? New recommendations (12)

Recommendations	Class
Recommendations for patients with AF and an ACS, PCI, or CCS	
Recommendations for AF patients with ACS	
In AF patients with ACS undergoing an uncomplicated PCI, early cessation (≤1 week) of aspirin and continuation of dual therapy with an OAC and a $P2Y_{12}$ inhibitor (preferably clopidogrel) for up to 12 months is recommended if the risk of stent thrombosis is low or if concerns about bleeding risk prevail over concerns about risk of stent thrombosis, irrespective of the type of stent used.	I
Recommendations in AF patients with a CCS undergoing PCI	
After uncomplicated PCI, early cessation (≤1 week) of aspirin and continuation of dual therapy with OAC for up to 6 months and clopidogrel is recommended if the risk of stent thrombosis is low or if concerns about bleeding risk prevail over concerns about risk of stent thrombosis, irrespective of the type of stent used.	I

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What is new in the 2020 Guidelines? New recommendations (13)

Recommendations	Class
Recommendations for the management of active bleeding on OAC	
Four-factor prothrombin complex concentrates should be considered in AF patients on VKA who develop a severe bleeding complication.	IIa
Recommendations for the management of AF during pregnancy	
Acute management	
In pregnant women with HCM, cardioversion should be considered for persistent AF.	IIa
Ibutilide or flecainide i.v. may be considered for termination of AF in stable patients with structurally normal hearts.	IIb

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What is new in the 2020 Guidelines? New recommendations (14)

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Recommendations	Class
Recommendations for the management of AF during pregnancy	
Long-term management (oral administration of drugs)	
Flecainide, propafenone, or sotalol should be considered to prevent AF if atrioventricular nodal-blocking drugs fail.	IIa
Digoxin or verapamil should be considered for rate control if beta-blockers fail.	IIa
Recommendations for postoperative AF	
Long-term OAC therapy to prevent thromboembolic events should be considered in patients at risk for stroke with postoperative AF after non-cardiac surgery, considering the anticipated net clinical benefit of OAC and informed patient preferences.	IIa
Beta-blockers should not be used routinely for the prevention of postoperative AF in patients undergoing non-cardiac surgery.	III

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What is new in the 2020 Guidelines? New recommendations (15)

Recommendations	Class
Recommendations pertaining to sex-related differences in AF	
Women with symptomatic paroxysmal or persistent AF should be offered timely access to rhythm control therapies, including AF catheter ablation, when appropriate for medical reasons.	**IIa**
Recommendations for quality measures in AF	
The introduction of tools to measure quality of care and identify opportunities for improved treatment quality and AF patient outcome should be considered by practitioners and institutions.	**IIa**

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Changes in the recommendations (1)

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Recommendations about integrated AF management			
2020	**Class**	**2016**	**Class**
To optimize shared decision-making about specific AF treatment option(s) in consideration, it is recommended that: • Physicians inform the patient about advantages/limitations and benefit/risks associated with considered treatment option(s); and • Discuss the potential burden of the treatment with the patient and include the patient's perception of treatment burden into the treatment decision.	I	Placing patients in a central role in decision-making should be considered in order to tailor management to patient preferences and improve adherence to long-term therapy	IIa

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Changes in the recommendations (2)

Recommendations for the prevention of thromboembolic events in AF			
2020	**Class**	**2016**	**Class**
For bleeding risk assessment, a formal structured risk-score–based bleeding risk assessment is recommended to help identify non-modifiable and address modifiable bleeding risk factors in all AF patients, and to identify patients potentially at high risk of bleeding who should be scheduled for early and more frequent clinical review and follow-up.	I	Bleeding risk scores should be considered in AF patients on oral anticoagulation to identify modifiable risk factors for major bleeding.	IIa

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Changes in the recommendations (3)

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Recommendations for the prevention of thromboembolic events in AF (continued)			
2020	**Class**	**2016**	**Class**
In patients on VKAs with low time in INR therapeutic range (e.g. TTR <70%), recommended options are: • Switching to a NOAC but ensuring good adherence and persistence with therapy; or	**I**	AF patients already on treatment with a VKAs may be considered for NOAC treatment if TTR is not well controlled despite good adherence, or if patient preference without contraindications to NOAC (e.g. prosthetic valve).	**IIb**
• Efforts to improve TTR (e.g. education/counselling and more frequent INR checks).	**IIa**		

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Changes in the recommendations (4)

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Recommendations for rhythm control/catheter ablation of AF			
AF catheter ablation after drug therapy failure			
2020	**Class**	**2016**	**Class**
AF catheter ablation for PVI is recommended for rhythm control after one failed or intolerant class I or III AAD, to improve symptoms of AF recurrences in patients with: • Paroxysmal AF, or • Persistent AF without major risk factors for AF recurrence, or • Persistent AF with major risk factors for AF recurrence.	I	Catheter or surgical ablation should be considered in patients with symptomatic persistent or long-standing persistent AF refractory to AAD therapy to improve symptoms, considering patient choice, benefit and risk, supported by an AF Heart Team.	IIa

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Changes in the recommendations (5)

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Recommendations for rhythm control/catheter ablation of AF (continued)			
First-line therapy			
2020	**Class**	**2016**	**Class**
AF catheter ablation: • Is recommended to reverse LV dysfunction in AF patients when tachycardia-induced cardiomyopathy is highly probable, independent of their symptom status.	**I**	AF ablation should be considered in symptomatic patients with AF and HFrEF to improve symptoms and cardiac function when tachycardiomyopathy is suspected.	**IIa**
• Should be considered in selected AF patients with HFrEF to improve survival and reduce HF hospitalization.	**IIa**		

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Changes in the recommendations (6)

Recommendations for rhythm control/catheter ablation of AF (continued)			
Techniques and technologies			
2020	**Class**	**2016**	**Class**
Complete electrical isolation of the pulmonary veins is recommended during all AF catheter-ablation procedures.	**I**	Catheter ablation should target isolation of the pulmonary veins using radiofrequency ablation or cryothermy balloon catheters.	**IIa**
If patient has a history of CTI-dependent atrial flutter or if typical atrial flutter is induced at the time of AF ablation, delivery of a CTI lesion may be considered.	**IIb**	Ablation of common atrial flutter should be considered to prevent recurrent flutter as part of an AF ablation procedure if documented or occurring during the AF ablation.	**IIa**

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Changes in the recommendations (7)

Recommendations for rhythm control/catheter ablation of AF (continued)			
Lifestyle modification and other strategies to improve outcomes of ablation			
2020	**Class**	**2016**	**Class**
Weight loss is recommended in obese patients with AF, particularly those who are being evaluated to undergo AF ablation.	I	In obese patients with AF, weight loss together with management of other risk factors should be considered to reduce AF burden and symptoms.	IIa
Recommendations for stroke risk management peri cardioversion			
In patients with AF undergoing cardioversion, NOACs are recommended with at least similar efficacy and safety as warfarin.	I	Anticoagulation with heparin or a NOAC should be initiated as soon as possible before every cardioversion of AF or atrial flutter.	IIa

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Changes in the recommendations (8)

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Recommendations for stroke risk management peri catheter ablation			
2020	**Class**	**2016**	**Class**
After AF catheter ablation, it is recommended that: • Systemic anticoagulation with warfarin or a NOAC is continued for at least 2 months post ablation, and • Long-term continuation of systemic anticoagulation beyond 2 months post ablation is based on the patient's stroke risk profile and not on the apparent success or failure of the ablation procedure.	**I**	All patients should receive oral anticoagulation for at least 8 weeks after catheter ablation.	**IIa**

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Changes in the recommendations (9)

Recommendations for long-term antiarrhythmic drugs			
2020	**Class**	**2016**	**Class**
Amiodarone is recommended for long-term rhythm control in all AF patients, including those with HFrEF. However, owing to its extracardiac toxicity, other AADs should be considered first whenever possible.	**I**	Amiodarone is more effective in preventing AF recurrences than other AAD, but extracardiac toxic effects are common and increase with time. For this reason, other AAD should be considered first.	**IIa**

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Changes in the recommendations (10)

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Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with AF			
2020	**Class**	**2016**	**Class**
Attention to good BP control is recommended in AF patients with hypertension to reduce AF recurrences and risk of stroke and bleeding.	**I**	Blood pressure control in anticoagulated patients with hypertension should be considered to reduce the risk of bleeding	**IIa**
Physical activity should be considered to help prevent AF incidence or recurrence, with the exception of excessive endurance exercise, which may promote AF.	**IIa**	Moderate regular physical activity is recommended to prevent AF, while athletes should be counselled that long-lasting intense sports participation can promote AF	**I**

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Changes in the recommendations (11)

Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with AF (continued)			
2020	**Class**	**2016**	**Class**
Optimal management of OSA may be considered, to reduce AF incidence, AF progression, AF recurrences, and symptoms.	**IIb**	OSA treatment should be optimized to reduce AF recurrences and improve AF treatment results.	**IIa**

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Changes in the recommendations (12)

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Recommendations for stroke prevention in AF patients after ICH			
2020	**Class**	**2016**	**Class**
In AF patients at high risk of ischaemic stroke, (re-)initiation of OAC, with preference for NOACs over VKAs in NOAC-eligible patients, should be considered in consultation with a neurologist/stroke specialist after: • A trauma-related ICH • Acute spontaneous ICH (which includes subdural, subarachnoid, or intracerebral haemorrhage), after careful consideration of risks and benefits.	**IIa**	After ICH, oral anticoagulation in patients with AF may be reinitiated after 4–8 weeks provided the cause of bleeding or the relevant risk factor has been treated or controlled.	**IIb**

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Changes in the recommendations (13)

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Recommendations for postoperative AF			
2020	**Class**	**2016**	**Class**
Long-term OAC therapy to prevent thromboembolic events may be considered in patients at risk for stroke with postoperative AF after cardiac surgery, considering the anticipated net clinical benefit of OAC therapy and informed patient preferences.	**IIb**	Long-term anticoagulation should be considered in patients with AF after cardiac surgery at risk for stroke, considering individual stroke and bleeding risk.	**IIa**

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Table 3 Definition of atrial fibrillation (1)

	Definition
AF	A supraventricular tachyarrhythmia with uncoordinated atrial electrical activation and consequently ineffective atrial contraction. *Electrocardiographic characteristics of AF include:* • Irregularly irregular R-R intervals (when atrioventricular conduction is not impaired), • Absence of distinct repeating P waves, and • Irregular atrial activations.
	Currently used terms
Clinical AF	*Symptomatic* or *asymptomatic* AF that is documented by surface ECG. The minimum duration of an ECG tracing of AF required to establish the diagnosis of clinical AF is at least 30 seconds, or entire 12-lead ECG.

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Table 3 Definition of atrial fibrillation (2)

	Currently used terms (continued)
Atrial high rate episode (AHRE), subclinical AF	Refers to individuals *without symptoms attributable to AF,* in whom *clinical AF is NOT previously detected (that is, there is no surface ECG tracing of AF).* **AHRE** – events fulfilling programmed or specified criteria for AHRE that are detected by CIEDs with an atrial lead allowing automated continuous monitoring of atrial rhythm and tracings storage. CIED-recorded AHRE need to be visually inspected because some AHRE may be electrical artefacts/false positives. **Subclinical AF** includes AHRE confirmed to be AF, AFL, or an AT, or AF episodes detected by insertable cardiac monitor or wearable monitor and confirmed by visually reviewed intracardiac electrograms or ECG-recorded rhythm.

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Recommendations for diagnosis of AF

Recommendations	Class	Level
ECG documentation is required to establish the diagnosis of AF. • A standard 12-lead ECG recording or a single-lead ECG tracing of ≥30 seconds showing heart rhythm with no discernible repeating P waves and irregular RR intervals (when atrioventricular conduction is not impaired) is diagnostic of clinical AF.	I	B

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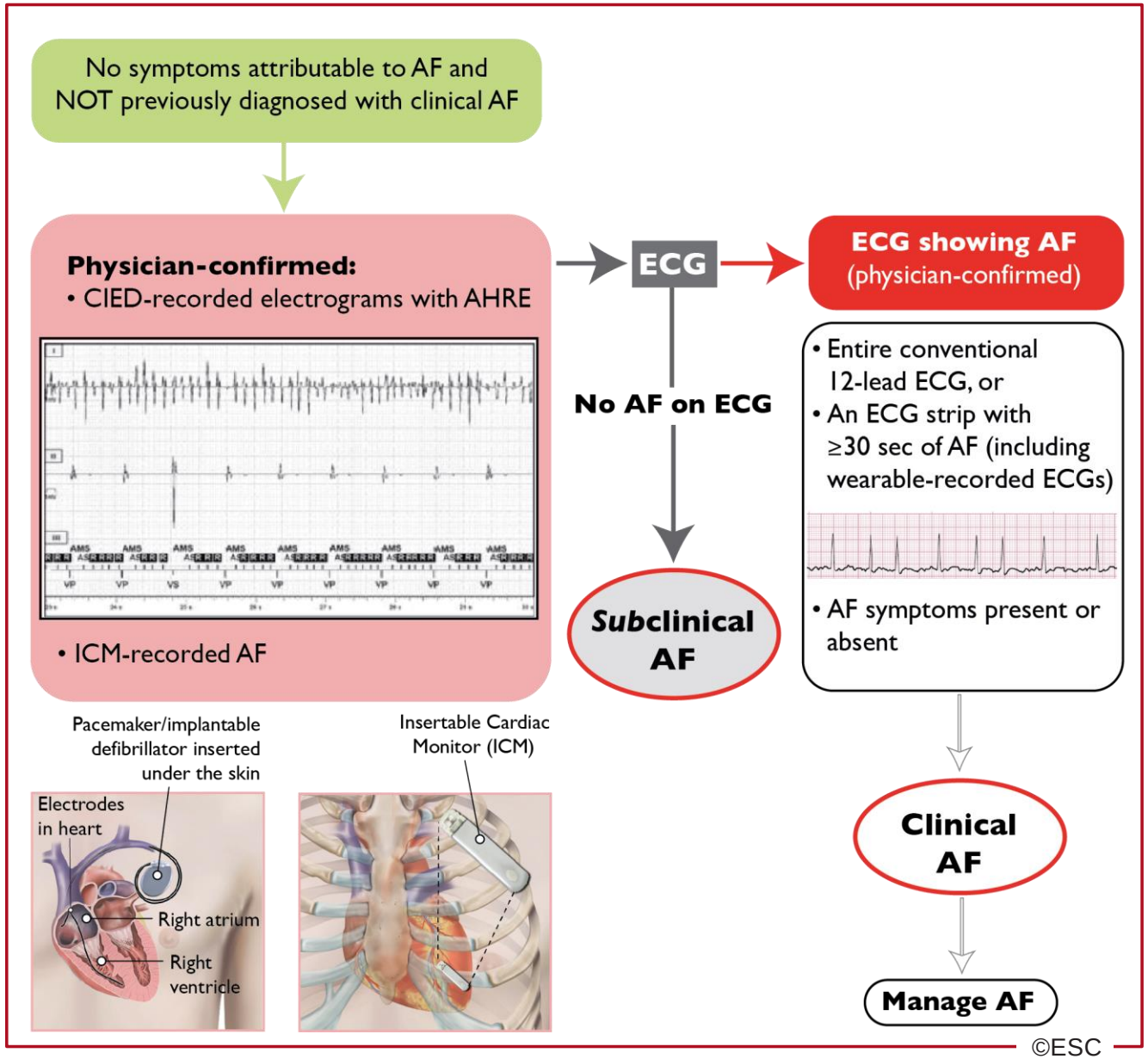
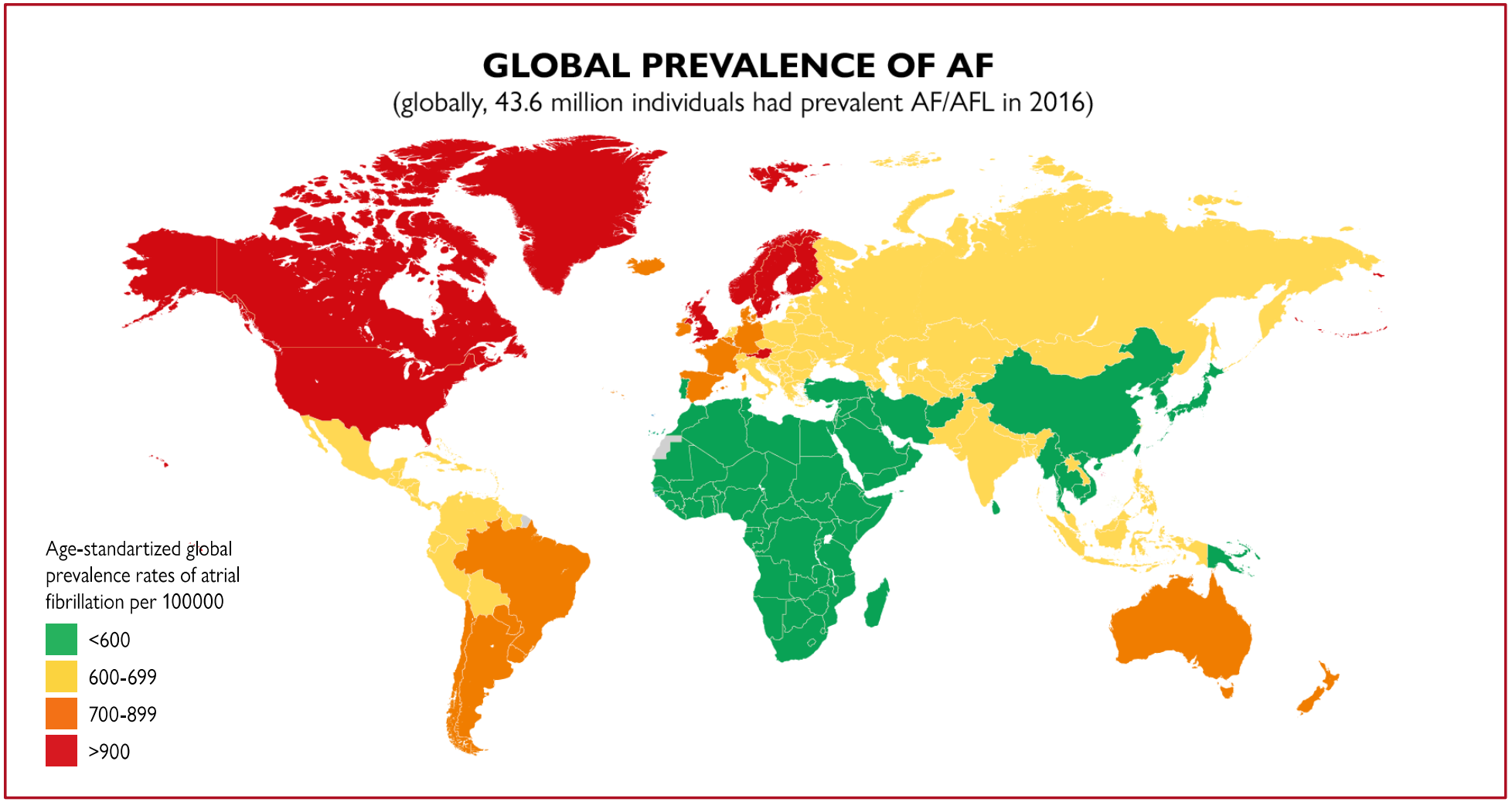


Figure 1 Diagnosis of AHRE/subclinical atrial fibrillation

CIEDs with an atrial lead can monitor atrial rhythm and store the tracings. ICM have no intra-cardiac leads but continuously monitor cardiac electrical activity by recording and analysing a single-lead bipolar surface ECG based on specific algorithm.
Left-bottom image: pacemaker with a right atrial lead, and a ventricular lead in the right ventricular apex. In addition to pacing at either site, these leads can sense activity in the respective cardiac chamber. The device can also detect pre-programmed events, such as AHRE.
Right-bottom image: subcutaneous ICM: these devices have no intra-cardiac leads and essentially record a single, bipolar, surface ECG with inbuilt algorithms for detection of AHRE or AF.

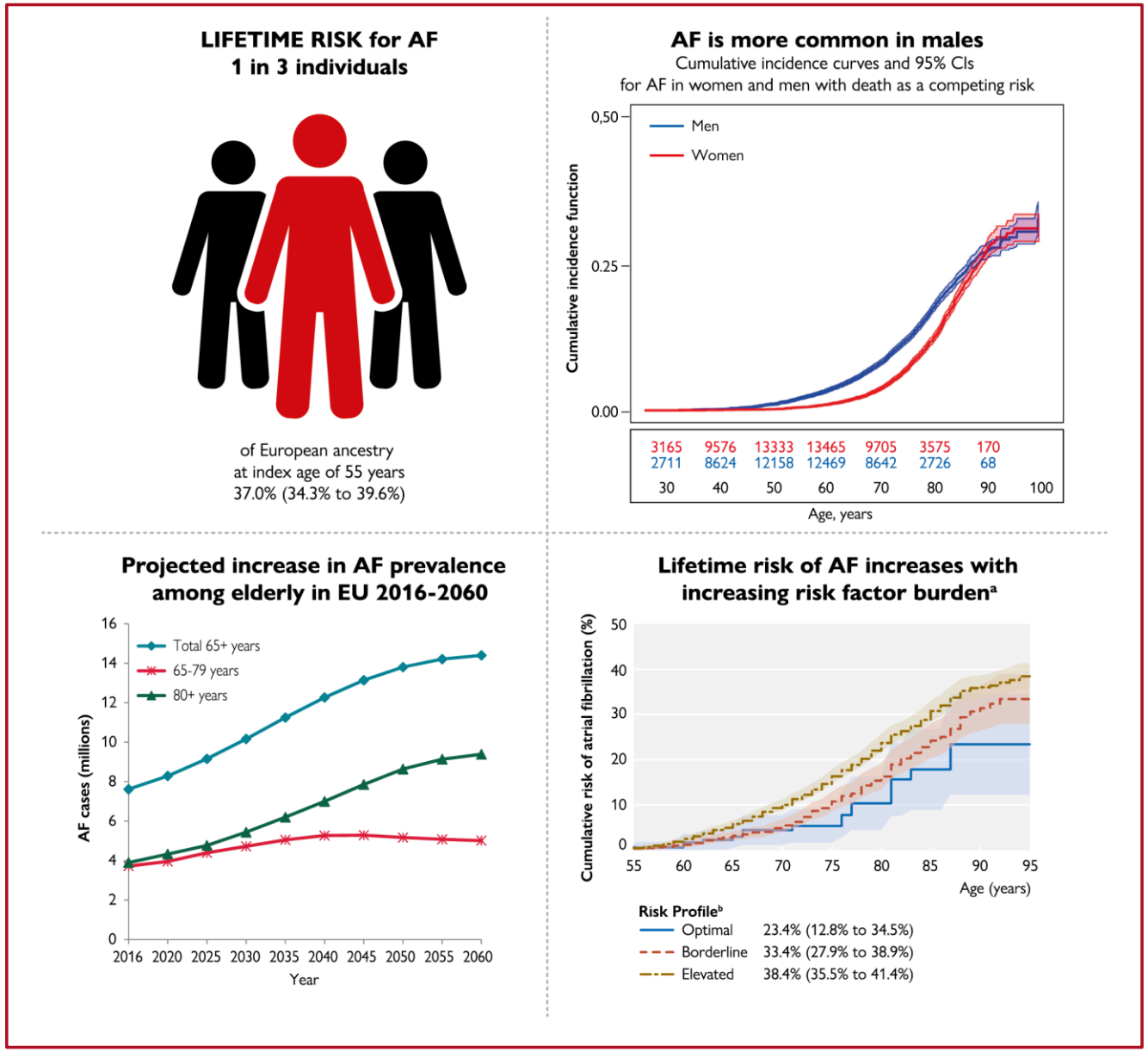
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Figure 2 (1) Epidemiology of AF: prevalence



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Figure 2 (2)

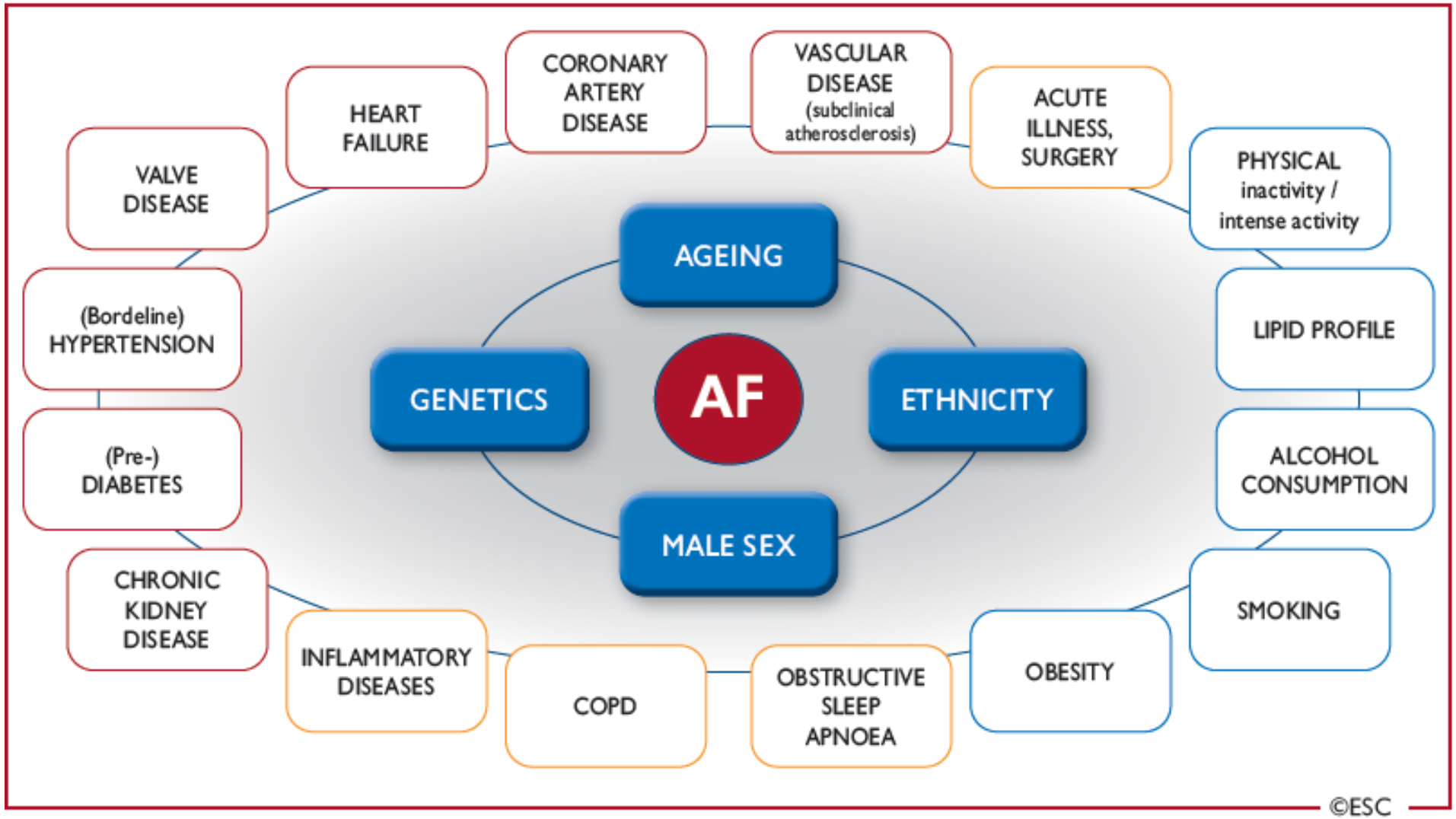
Epidemiology of AF: lifetime risk and projected rise in the incidence and prevalence

[a]Smoking, alcohol consumption, body mass index, BP, diabetes mellitus (type 1 or 2), and history of myocardial infarction or heart failure. [b]Risk profile: *optimal* – all risk factors are negative or within the normal range; *borderline* – no elevated risk factors but >1 borderline risk factor; *elevated* – >1 elevated risk factor.

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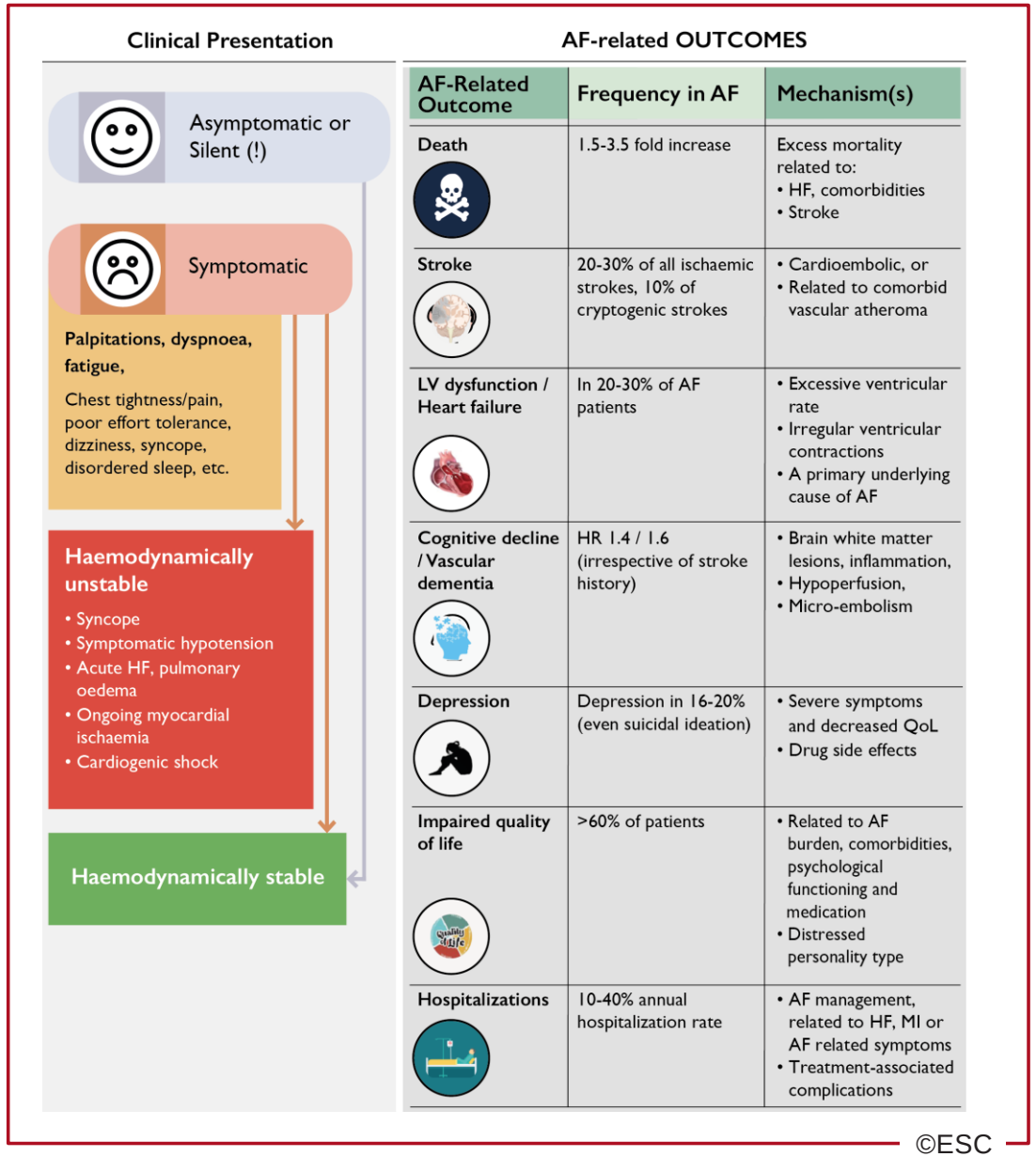
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Figure 3 Summary of risk factors for incident AF



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Figure 4 Clinical presentation of AF and AF-related outcomes

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Table 4 Classification of AF (1)

AF pattern	Definition
First diagnosed	AF not diagnosed before, irrespective of its duration or the presence/severity of AF-related symptoms.
Paroxysmal	AF that terminates spontaneously or with intervention within 7 days of onset.
Persistent	AF that is continuously sustained beyond 7 days, including episodes that are terminated by cardioversion (drugs or direct current cardioversion) after 7 days or more.
Long-standing persistent	Continuous AF of >12 months' duration when decided to adopt a rhythm control strategy.
Permanent	AF that is accepted by the patient and physician, and no further attempts to restore/maintain sinus rhythm will be undertaken. Permanent AF represents a therapeutic attitude of the patient and physician rather than an inherent pathophysiological attribute of AF, and the term should not be used in the context of a rhythm control strategy with antiarrhythmic drug therapy or AF ablation. Should a rhythm control strategy be adopted, the arrhythmia would be re-classified as 'long-standing persistent AF'.

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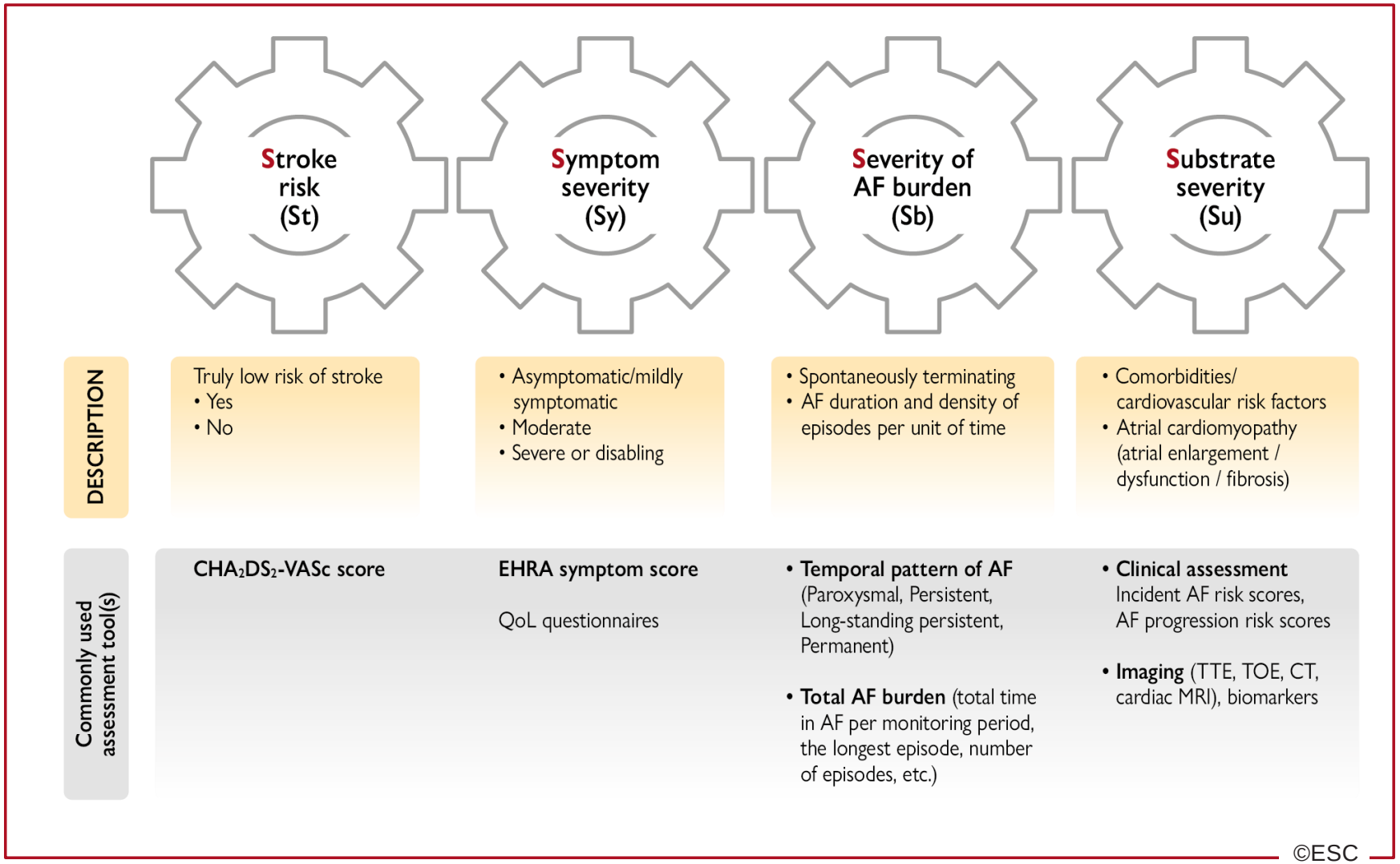
Table 4 Classification of AF (2)

Terminology that should be abandoned	
AF pattern	**Definition**
Lone AF	A historical descriptor. Increasing knowledge about the pathophysiology of AF shows that in every patient a cause is present. Hence, this term is potentially confusing and should be abandoned.
Valvular/non-valvular AF	Differentiates patients with moderate/severe mitral stenosis and those with mechanical prosthetic heart valve(s) from other patients with AF, but may be confusing and should not be used.
Chronic AF	Has variable definitions and should not be used to describe populations of AF patients.

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Figure 5 4S-AF scheme as an example of structured characterization of AF



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Recommendations for structured characterization of AF

Recommendations	Class	Level
Structured characterization of AF, which includes clinical assessment of stroke risk, symptom status, burden of AF, and evaluation of substrate, should be considered in all AF patients, to streamline the assessment of AF patients at different healthcare levels, inform treatment decision-making, and facilitate optimal management of AF patients.	IIa	C

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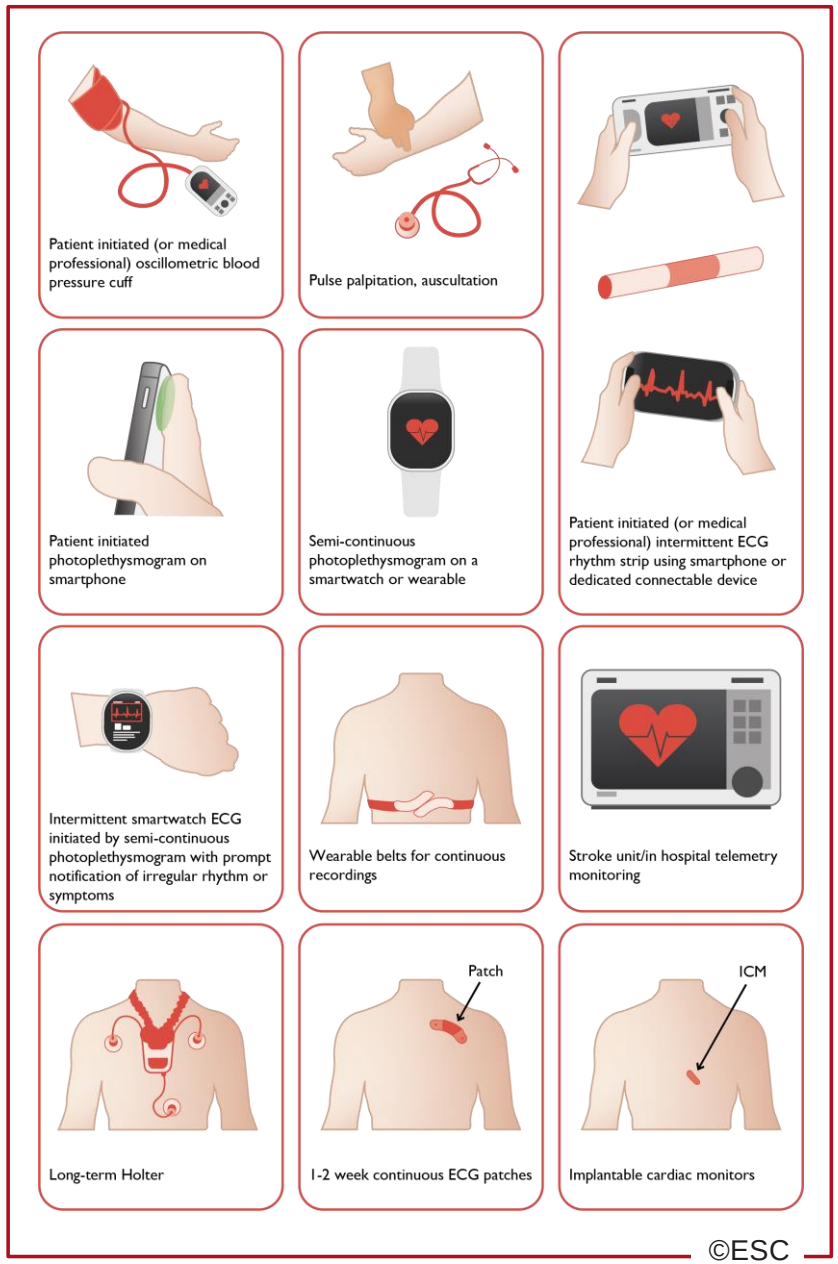


Figure 6 Systems used for AF screening

Pulse palpation, automated BP monitors, single-lead ECG devices, PPG devices, other sensors (using seismocardiography, accelerometers, and gyroscopes, etc.) used in applications for smartphones, wrist bands, and watches. Intermittent smartwatch detection through PPG or ECG recordings. Smartwatches and other 'wearables' can passively measure pulse rate from the wrist using an optical sensor for PPG and alerting the consumer of a pulse irregularity (based on a specific algorithm for AF detection analysing pulse irregularity and variability

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Table 5 Sensitivity and specificity of various AF screening tools considering the 12-lead ECG as the gold standard

	Sensitivity	Specificity
Pulse taking	87–97%	70–81%
Automated BP monitors	93–100%	86–92%
Single lead ECG	94–98%	76–95%
Smartphone apps	91.5–98.5%	91.4–100%
Watches	97–99%	83–94%

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Figure 7 Potential benefits from and risks of screening for AF

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AF SCREENING

RISKS

- Abnormal results may cause anxiety
- ECG misinterpretation results may lead to overdiagnosis and overtreatment
- ECG may detect other abnormalities (true or false positives) that may lead to invasive tests and treatments that have the potential for serious harm (e.g., angiography / revascularisation with bleeding, contrast-induced nephropathy and allergic reactions to the contrast)

BENEFITS

Prevention of:
- Stroke/SE using OAC in patients at risk
- Subsequent onset of symptoms

Prevention/reversal of:
- Electrical/mechanical atrial remodelling
- AF-related haemodynamic derangements
- Atrial and ventricular tachycardia-induced cardiopmyopathy

Prevention/reduction of:
- AF-related morbidity; hospitalization; mortality

Reduction of:
- The outcomes associated with conditions / diseases associated with AF that are discovered and treated as a consequence of the examinations prompted by AF detection

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Recommendations for screening to detect AF (1)

Recommendations	Class	Level
Opportunistic screening for AF by pulse taking or ECG rhythm strip is recommended in patients ≥65 years of age.	I	B
It is recommended to interrogate pacemakers and implantable cardioverter defibrillators on a regular basis for AHRE.[a]	I	B

[a]See *sections* for diagnostic criteria for AF and AHRE, and for the management of patients with AHRE.

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Recommendations for screening to detect AF (2)

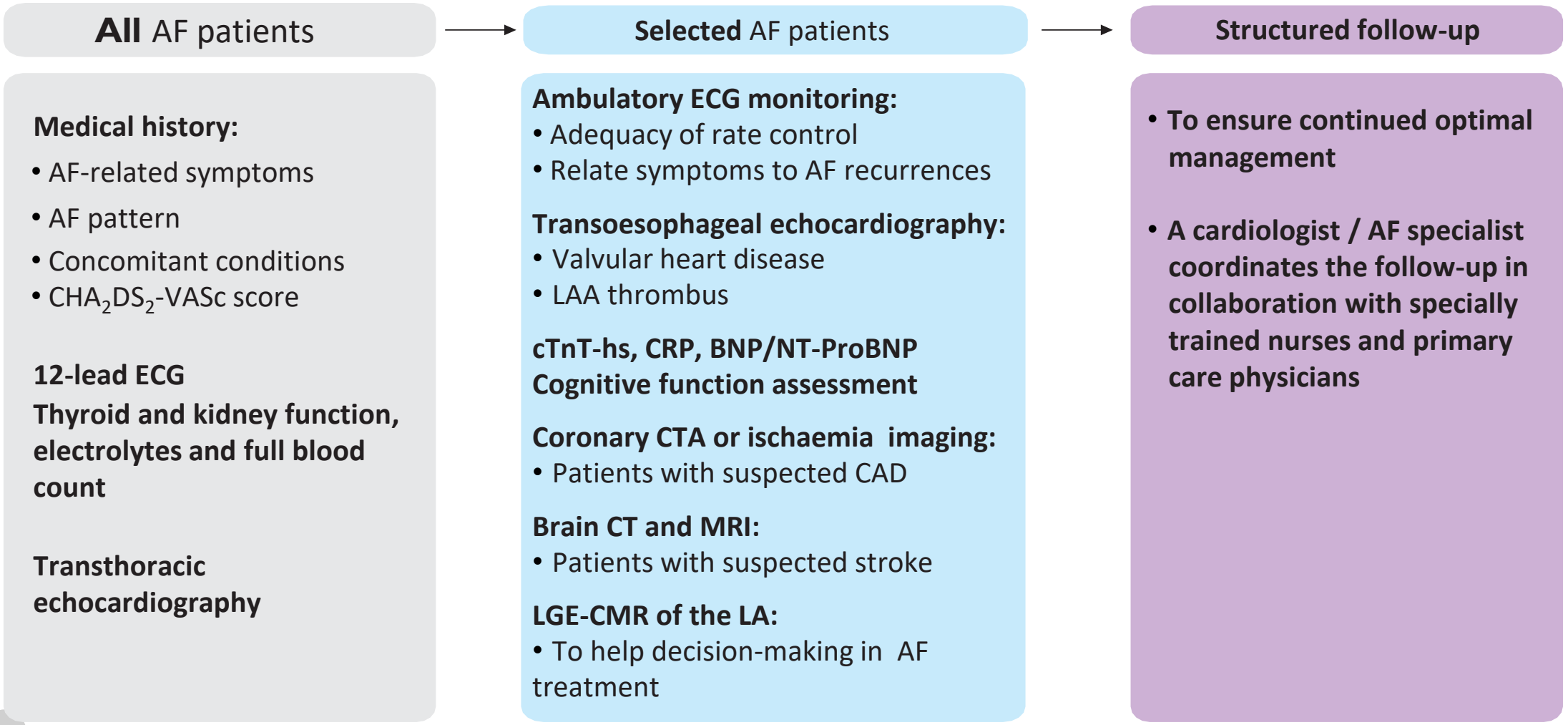
Recommendations	Class	Level
When screening for AF it is recommended that: • The individuals undergoing screening are informed about the significance and treatment implications of detecting AF. • A structured referral platform is organized for screen-positive cases for further physician-led clinical evaluation to confirm the diagnosis of AF and provide optimal management of patients with confirmed AF. • Definite diagnosis of AF in screen-positive cases is established only after physician reviews the single-lead ECG recording of ≥30 seconds or 12-lead ECG and confirms that it shows AF.	I	B
Systematic ECG screening should be considered to detect AF in individuals aged ≥75 years, or those at high risk of stroke.	IIa	B

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Figure 8 Diagnostic work-up and follow-up in AF patients

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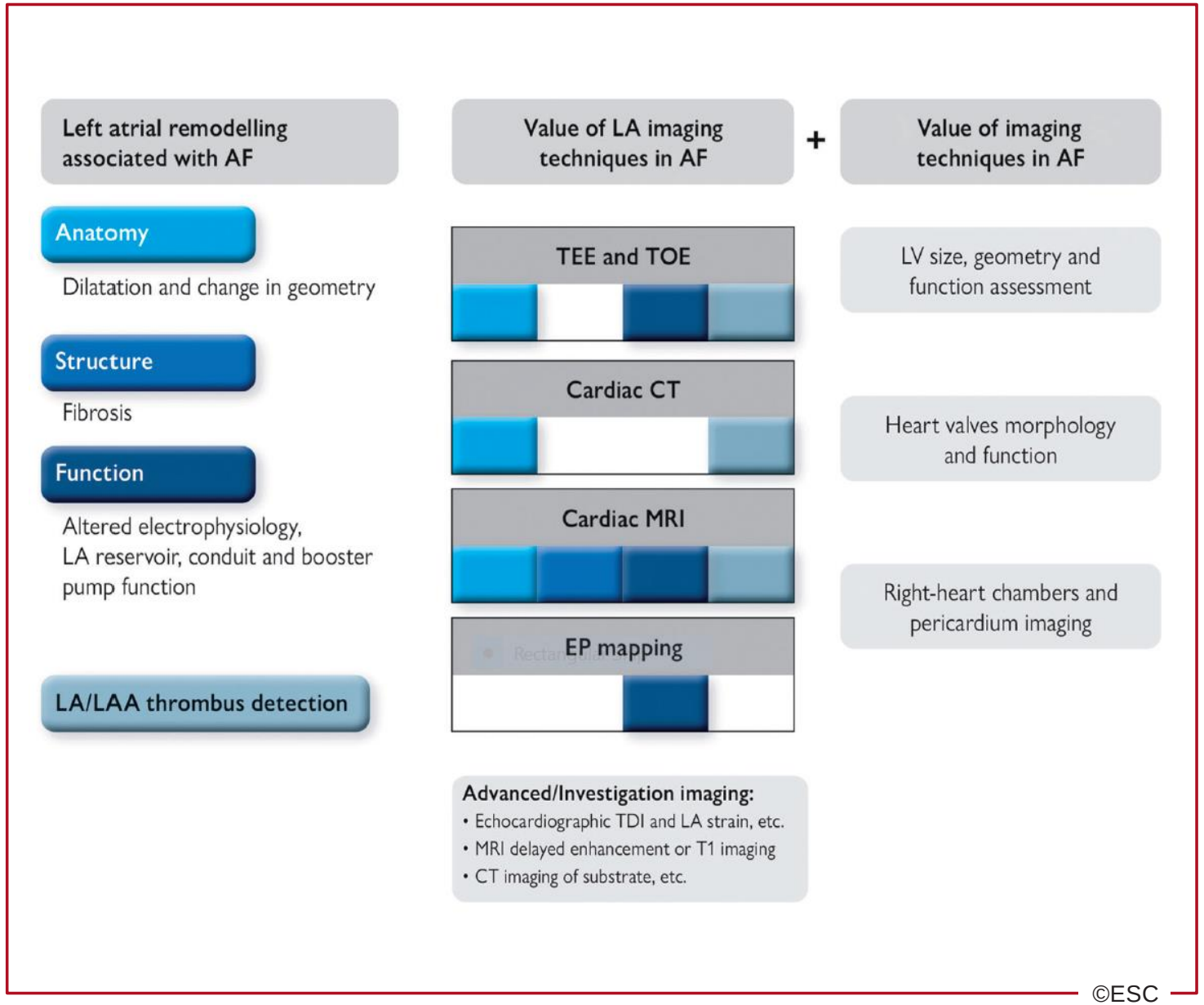
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Table 6 EHRA symptom scale

Score	Symptoms	Description
1	None	AF does not cause any symptoms
2a	Mild	Normal daily activity not affected by symptoms related to AF
2b	Moderate	Normal daily activity not affected by symptoms related to AF, but patient troubled by symptoms
3	Severe	Normal daily activity affected by symptoms related to AF
4	Disabling	Normal daily activity discontinued

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Figure 9
Imaging in AF

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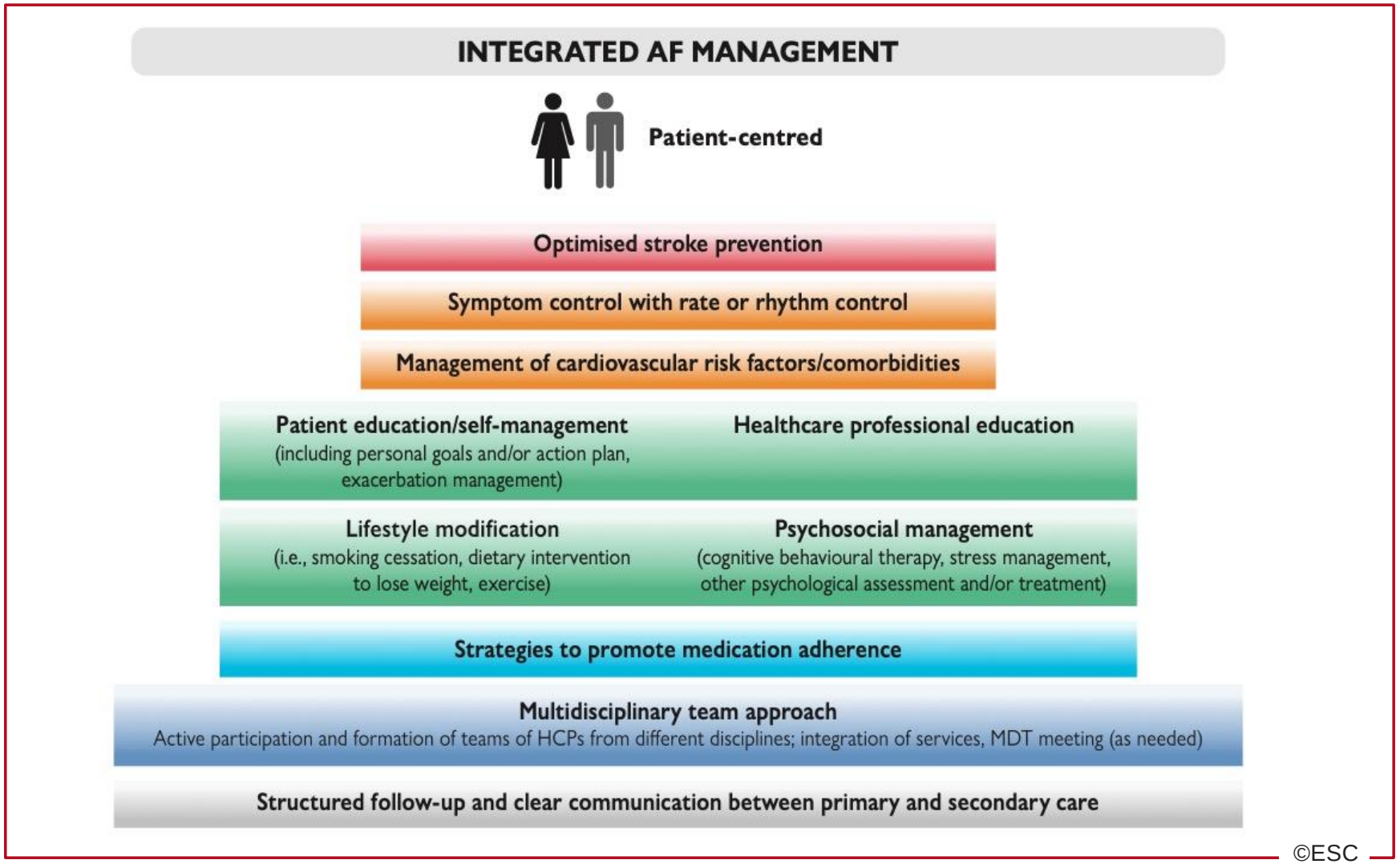
Recommendations for diagnostic evaluation of patients with AF

Recommendations	Class	Level
In patients with AF, it is recommended to: • Evaluate AF-related symptoms (including fatigue, tiredness, exertional shortness of breath, palpitations, and chest pain) and quantify the patient symptom status using the modified EHRA symptom scale before and after initiation of treatment. • Evaluate AF-related symptoms before and after cardioversion of persistent AF to aid rhythm control treatment decisions.	I	C

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Figure 10 Components of integrated AF management



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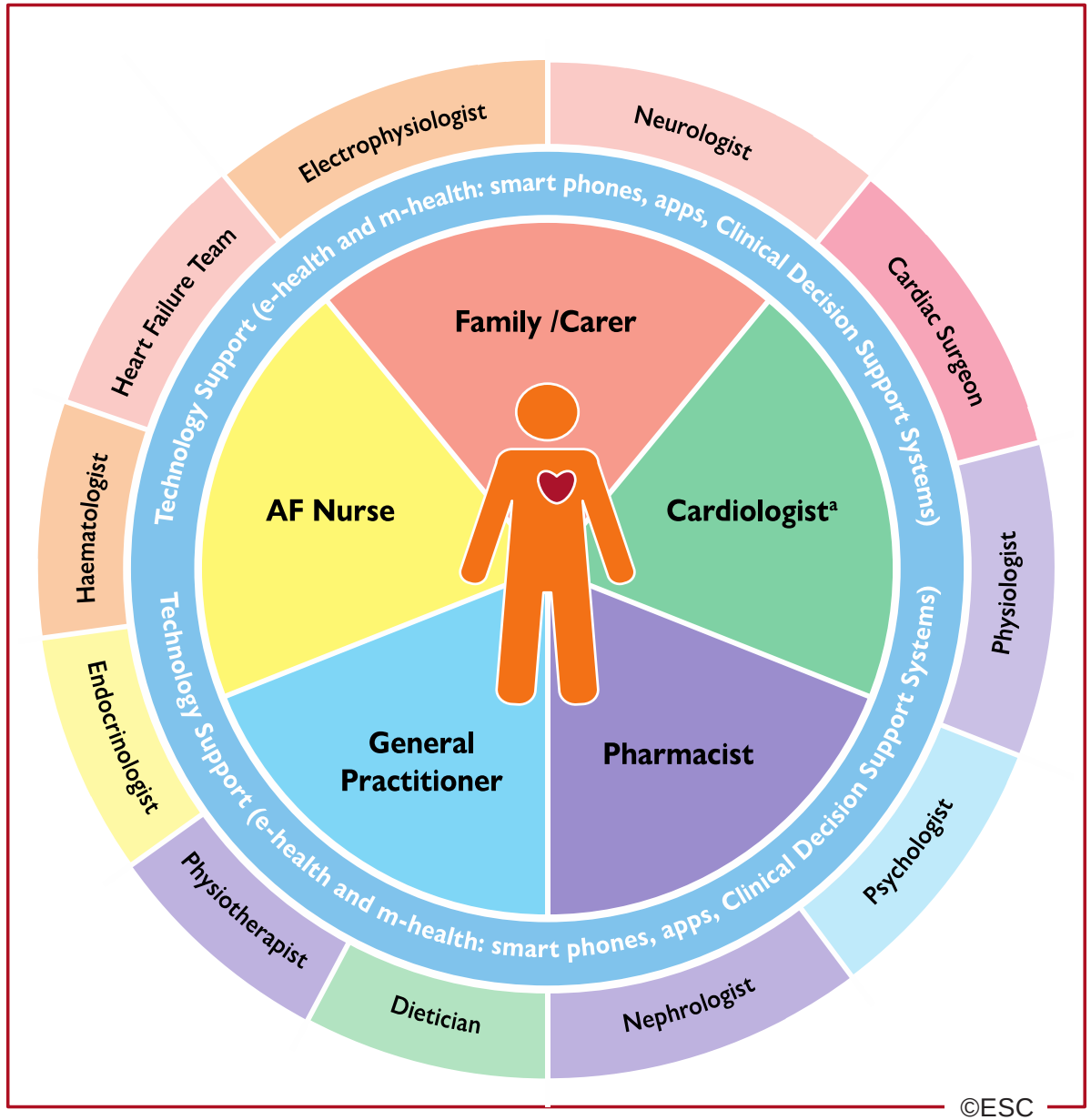


Figure 11
Integrated AF management team (an example)

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Recommendations about integrated AF management

Recommendations	Class	Level
To optimize shared decision-making about specific AF treatment option(s) in consideration, it is recommended that physicians: • Inform the patient about the advantages/limitations and benefit/risks associated with the treatment option(s) being considered; and • Discuss the potential burden of the treatment with the patient and include the patient's perception of treatment burden into the treatment decision.	I	C
It is recommended to routinely collect PROs to measure treatment success and improve patient care.	I	C
Integrated management with a structured multidisciplinary approach including healthcare professionals, patients, and their family/carers, should be used in all AF patients to improve clinical outcomes.	IIa	B

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Table 7 Stroke risk factors in patients with AF

Most commonly studied clinical risk factors (a systematic review)	Positive studies/All studies	Other clinical risk factors	Imaging biomarkers	Blood/urine biomarkers
Stroke/TIA/systemic embolism	15/16	Impaired renal function/CKD	*Echocardiography*	Cardiac troponin T and I Natriuretic peptides Cystatin C Proteinuria CrCl/eGFR CRP IL-6 GDF-15 von Willebrand factor D-dimer
Hypertension	11/20	OSA	LA dilatation Spontaneous contrast or thrombus in LA Low LAA velocities Complex aortic plaque	
Ageing (per decade)	9/13	Hypertrophic cardiomyopathy		
Structural heart disease	9/13	Amyloidosis in degenerative cerebral and heart diseases		
Diabetes mellitus	9/14	Hyperlipidaemia		
Vascular disease	6/17	Smoking	*Cerebral imaging*	
CHF/LV dysfunction	7/18	Metabolic syndrome	Small-vessel disease	
Sex category (female)	8/22	Malignancy		

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Table 8 CHA_2DS_2-VASc score (1)

CHA_2DS_2-VASc score			
Risk factors and definitions		**Points awarded**	**Comment**
C	**Congestive heart failure** Clinical HF, or objective evidence of moderate to severe LV dysfunction, or HCM	**1**	Recent decompensated HF irrespective of LVEF (thus incorporating HFrEF or HFpEF), or the presence (even if asymptomatic) of moderate-severe LV systolic impairment on cardiac imaging; HCM confers a high stroke risk and OAC is beneficial for stroke reduction.
H	**Hypertension** or on antihypertensive therapy	**1**	History of hypertension may result in vascular changes that predispose to stroke, and a well-controlled BP today may not be well-controlled over time. Uncontrolled BP − the optimal BP target associated with the lowest risk of ischaemic stroke, death, and other cardiovascular outcomes is 120−129/<80 mmHg.
A	**Age 75 years or older**	**2**	Age is a powerful driver of stroke risk, and most population cohorts show that the risk rises from age 65 years upwards. Age-related risk is a continuum, but for reasons of simplicity and practicality, 1 point is given for age 65−74 years and 2 points for age ≥75 years.

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Table 8 CHA_2DS_2-VASc score (2)

CHA_2DS_2-VASc score			
Risk factors and definitions		**Points awarded**	**Comment**
D	**Diabetes mellitus** Treatment with oral hypoglycaemic drugs and/or insulin or fasting blood glucose >125 mg/dL (7 mmol/L)	**1**	Diabetes mellitus is a well-established risk factor for stroke, and more recently stroke risk has been related to duration of diabetes mellitus (the longer the duration of diabetes mellitus, the higher the risk of thromboembolism) and presence of diabetic target organ damage, e.g. retinopathy. Both type 1 and type 2 diabetes mellitus confer broadly similar thromboembolic risk in AF, although the risk may be slightly higher in patients aged <65 years with type 2 diabetes mellitus compared to patients with type 1 diabetes mellitus.
S	**Stroke** Previous stroke, TIA, or thromboembolism	**2**	Previous stroke, systemic embolism, or TIA confers a particularly high risk of ischaemic stroke, hence weighted 2 points. Although excluded from RCTs, AF patients with ICH (including haemorrhagic stroke) are at very high risk of subsequent ischaemic stroke, and recent observational studies suggest that such patients would benefit from oral anticoagulation.

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Table 8 CHA_2DS_2-VASc score (3)

CHA_2DS_2-VASc score			
Risk factors and definitions		**Points awarded**	**Comment**
V	**Vascular disease** Angiographically significant CAD, previous myocardial infarction, PAD, or aortic plaque	**1**	Vascular disease (PAD or myocardial infarction) confers a 17–22% excess risk, particularly in Asian patients. Angiographically significant CAD is also an independent risk factor for ischaemic stroke among AF patients (adjusted incidence rate ratio 1.29, 95% CI 1.08–1.53). Complex aortic plaque on the descending aorta, as an indicator of significant vascular disease, is also a strong predictor of ischaemic stroke.
A	**Age 65–74 years**	**1**	See above. Recent data from Asia suggest that the risk of stroke may rise from age 50–55 years upwards and that a modified CHA_2DS_2-VASc score may be used in Asian patients.
Sc	**Sex category (female)**	**1**	A stroke risk modifier rather than a risk factor.
Maximum score		**9**	

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Table 9 factors for bleeding with OAC and antiplatelet therapy

Non-modifiable	Potentially modifiable	Modifiable	Biomarkers
Age >65 years	Extreme frailty ± excessive risk of falls[a]	Hypertension/elevate SBP	GDF-15
Previous major bleeding	Anaemia	Concomitant antiplatelet/NSAID	Cystatin C / CKD-EPI
Severe renal impairment (on dialysis or renal transplant)	Reduced platelet count or function	Excessive alcohol intake	cTnT-hs
Severe hepatic dysfunction (cirrhosis)	Renal impairment with CrCl <60 mL/min	Non-adherence to OAC	Von Willebrand factor (+ other coagulation markers)
Malignancy	VKA management strategy[b]	Hazardous hobbies / occupations	
Genetic factors (e.g., CYP 2C9 polymorphisms)		Bridging therapy with heparin	
Previous stroke, small-vessel disease, etc.		INR control (target 2.0–3.0), target TTR >70%[c]	
Diabetes mellitus		Appropriate choice of OAC and correct dosing[d]	
Cognitive impairment/dementia			

[a]Walking aids; appropriate footwear; home review to remove trip hazards; neurological assessment where appropriate. [b]Increased INR monitoring, dedicated OAC clinicals, self-monitoring/self-management, educational/behavioural interventions. [c]For patients receiving VKA treatment. [d]Dose adaptation based on patient's age, body weight, and serum creatinine level.

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Table 10 Clinical risk factors in the HAS-BLED score (1)

Risk factors and definitions		Points awarded
H	**Uncontrolled hypertension** Systolic BP >160 mmHg	1
A	**Abnormal renal and/or hepatic function** Dialysis, transplant, serum creatinine >200 µmol/L, cirrhosis, bilirubin > × 2 upper limit of normal, AST/ALT/ALP >3 × upper limit of normal	1 point for each
S	**Stroke** Previous ischaemic or haemorrhagic[a] stroke	1
B	**Bleeding history or predisposition** Previous major haemorrhage or anaemia or severe thrombocytopenia	1

[a]Haemorrhagic stroke would also score 1 point under the 'B' criterion.

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Table 10 Clinical risk factors in the HAS-BLED score (2)

Risk factors and definitions		Points awarded
L	**Labile INR**[b] TTR <60% in patient receiving VKA	1
E	**Elderly** Aged >65 years or extreme frailty	1
D	**Drugs or excessive alcohol drinking** Concomitant use of antiplatelet or non-steroidal anti-inflammatory drugs; and/or excessive[c] alcohol per week	1 point for each
Maximum score		**9**

[b]Only relevant if patient receiving a VKA.
[c]Alcohol excess or abuse refers to a high intake (e.g. >14 units per week), where the clinician assesses there would be an impact on health or bleeding risk.

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Table 11 Dose selection criteria for NOACs

	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Standard dose	150 mg b.i.d.	20 mg o.d.	5 mg b.i.d.	60 mg o.d.
Lower dose	110 mg b.i.d.			
Reduced dose		15 mg o.d.	2.5 mg b.i.d.	30 mg o.d.
Dose-reduction criteria	Dabigatran 110 mg b.i.d. in patients with: • Age ≥80 years • Concomitant use of verapamil, or • Increased bleeding risk	CrCl 15–49 mL/min	At least 2 of 3 criteria: • Age ≥80 years, • Body weight ≤60 kg, or • Serum creatinine ≥1.5 mg/dL (133 µmol/L)	If any of the following: • CrCl 15–50 mL/min, • Body weight ≤60 kg, • Concomitant use of dronedarone, ciclosporin, erythromycin, or ketoconazole

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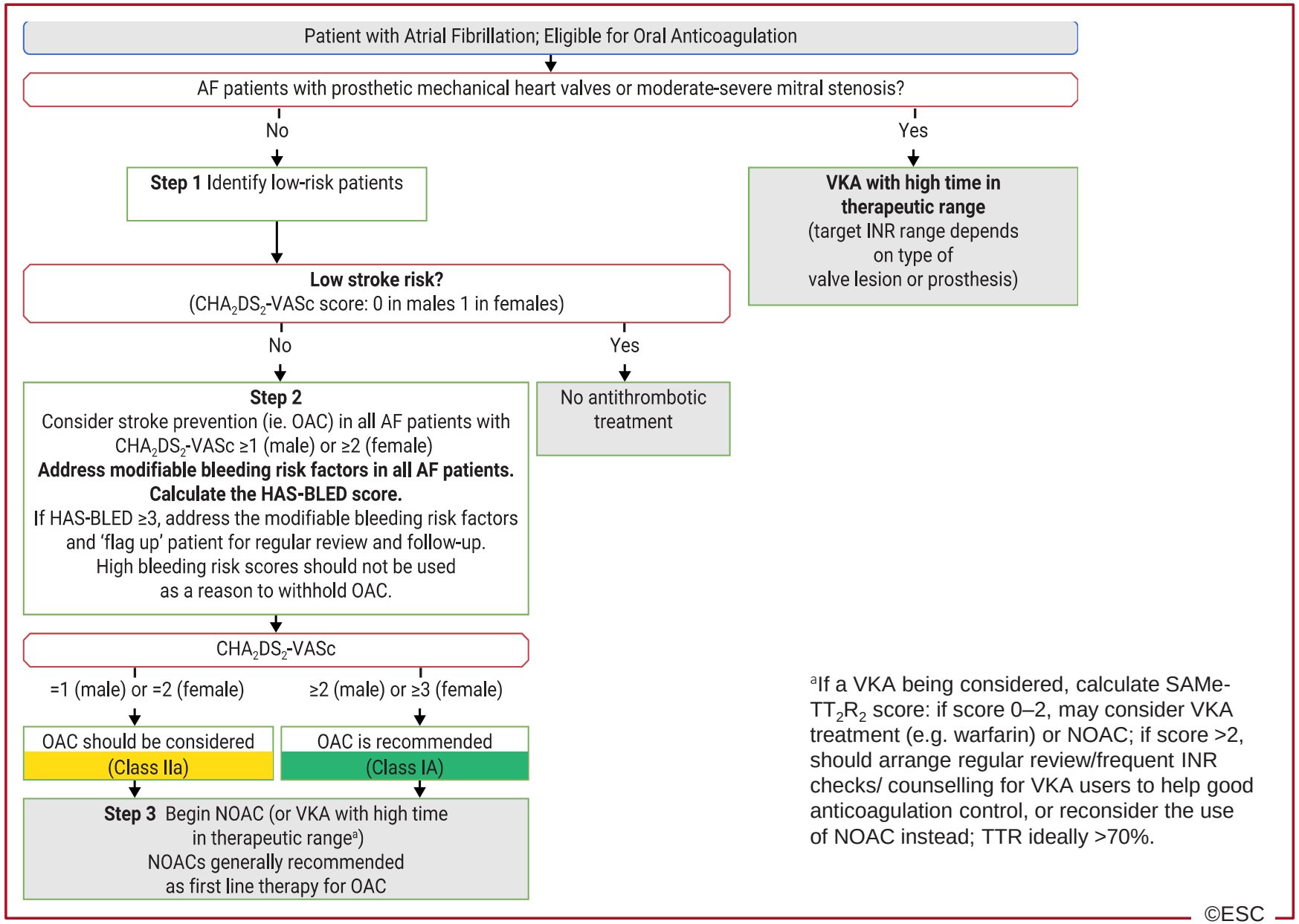
Table 12 Antithrombotic therapy after left atrial appendage occlusion

Device/patient	Aspirin	OAC	Clopidogrel	Comments
Watchman /low bleeding risk	75–325 mg/day indefinitely	Start warfarin after procedure (target INR 2–3) until 45 days or continue until adequate LAA sealing is confirmed[a] by TOE. NOAC is a possible alternative	Start 75 mg/day when OAC stopped, continue until 6 months after the procedure	Some centres do not withhold OAC at the time of procedure (no data to support/deny this approach)
Watchman /high bleeding risk	75–325 mg/day indefinitely	None	75 mg/day for 1–6 months while ensuring adequate LAA sealing[a]	Clopidogrel often given for shorter time in very high-risk situations
ACP/Amulet	75–325 mg/day indefinitely	None	75 mg/day for 1–6 months while ensuring adequate LAA sealing[a]	Clopidogrel may replace long-term aspirin if better tolerated

Note: Load aspirin or clopidogrel before procedure if untreated. Heparin with activated clotting time >250 seconds before or immediately after trans-septal punctures for all patients, followed by low-molecular-weight heparin when warfarin needed.

[a]Less than 5 mm leak.

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Patient with Atrial Fibrillation; Eligible for Oral Anticoagulation
AF patients with prosthetic mechanical heart valves or moderate-severe mitral stenosis?
No
Yes
Step 1 Identify low-risk patients
VKA with high time in therapeutic range (target INR range depends on type of valve lesion or prosthesis)
Low stroke risk? (CHA2DS2-VASc score: 0 in males 1 in females)
No
Yes
Step 2 Consider stroke prevention (ie. OAC) in all AF patients with CHA2DS2-VASc ≥1 (male) or ≥2 (female)
Address modifiable bleeding risk factors in all AF patients.
Calculate the HAS-BLED score.
If HAS-BLED ≥3, address the modifiable bleeding risk factors and 'flag up' patient for regular review and follow-up.
High bleeding risk scores should not be used as a reason to withhold OAC.
No antithrombotic treatment
CHA2DS2-VASc
=1 (male) or =2 (female)
≥2 (male) or ≥3 (female)
OAC should be considered (Class IIa)
OAC is recommended (Class IA)
Step 3 Begin NOAC (or VKA with high time in therapeutic range[a])
NOACs generally recommended as first line therapy for OAC
[a]If a VKA being considered, calculate SAMe-TT2R2 score: if score 0–2, may consider VKA treatment (e.g. warfarin) or NOAC; if score >2, should arrange regular review/frequent INR checks/ counselling for VKA users to help good anticoagulation control, or reconsider the use of NOAC instead; TTR ideally >70%.
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Recommendations for the prevention of thromboembolic events in AF (1)

Recommendations	Class	Level
For stroke prevention in AF patients who are eligible for OAC, NOACs are recommended in preference to VKAs (excluding patients with mechanical heart valves or moderate-to-severe mitral stenosis).	I	A
For stroke risk assessment, a risk-factor–based approach is recommended, using the CHA_2DS_2-VASc clinical stroke risk score to initially identify patients at 'low stroke risk' (CHA_2DS_2-VASc score = 0 in men, or 1 in women) who should not be offered antithrombotic therapy.	I	A
OAC is recommended for stroke prevention in AF patients with CHA_2DS_2-VASc score ≥2 in men or ≥3 in women.	I	A

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Recommendations for the prevention of thromboembolic events in AF (2)

Recommendations	Class	Level
OAC should be considered for stroke prevention in AF patients with a CHA_2DS_2-VASc score of 1 in men or 2 in women. Treatment should be individualized based on net clinical benefit and consideration of patient values and preferences.	IIa	B
For bleeding risk assessment, a formal structured risk-score–based bleeding risk assessment is recommended to help identify non-modifiable and address modifiable bleeding risk factors in all AF patients, and to identify patients potentially at high risk of bleeding who should be scheduled for early and more frequent clinical review and follow-up.	I	B
For a formal risk-score–based assessment of bleeding risk, the HAS-BLED score should be considered to help address modifiable bleeding risk factors, and to identify patients at high risk of bleeding (HAS-BLED score ≥3) for early and more frequent clinical review and follow-up.	IIa	B

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Recommendations for the prevention of thromboembolic events in AF (3)

Recommendations	Class	Level
Stroke and bleeding risk reassessment at periodic intervals is recommended to inform treatment decisions (e.g. initiation of OAC in patients no longer at low risk of stroke) and address potentially modifiable bleeding risk factors.[a]	I	B
In patients with AF initially at low risk of stroke, first reassessment of stroke risk should be made at 4–6 months after the index evaluation.	IIa	B
If a VKA is used, a target INR of 2.0–3.0 is recommended, with individual TTR ≥70%.	I	B

[a]Including uncontrolled BP; labile INRs (in a patient taking VKA); alcohol excess; concomitant use of NSAIDs or aspirin in an anticoagulated patient; bleeding tendency or predisposition (e.g. treat gastric ulcer, optimize renal or liver function etc.).

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Recommendations for the prevention of thromboembolic events in AF (4)

Recommendations	Class	Level
In patients on VKAs with low time in INR therapeutic range (e.g. TTR <70%), recommended options are: • Switching to a NOAC but ensuring good adherence and persistence with therapy; or	I	B
• Efforts to improve TTR (e.g. education/counselling and more frequent INR checks).	IIa	B
Antiplatelet therapy alone (monotherapy or aspirin in combination with clopidogrel) is not recommended for stroke prevention in AF.	III	A
Estimated bleeding risk, in the absence of absolute contraindications to OAC, should not in itself guide treatment decisions to use OAC for stroke prevention.	III	A
Clinical pattern of AF (i.e. first detected, paroxysmal, persistent, long-standing persistent, permanent) should not condition the indication to thromboprophylaxis.	III	B

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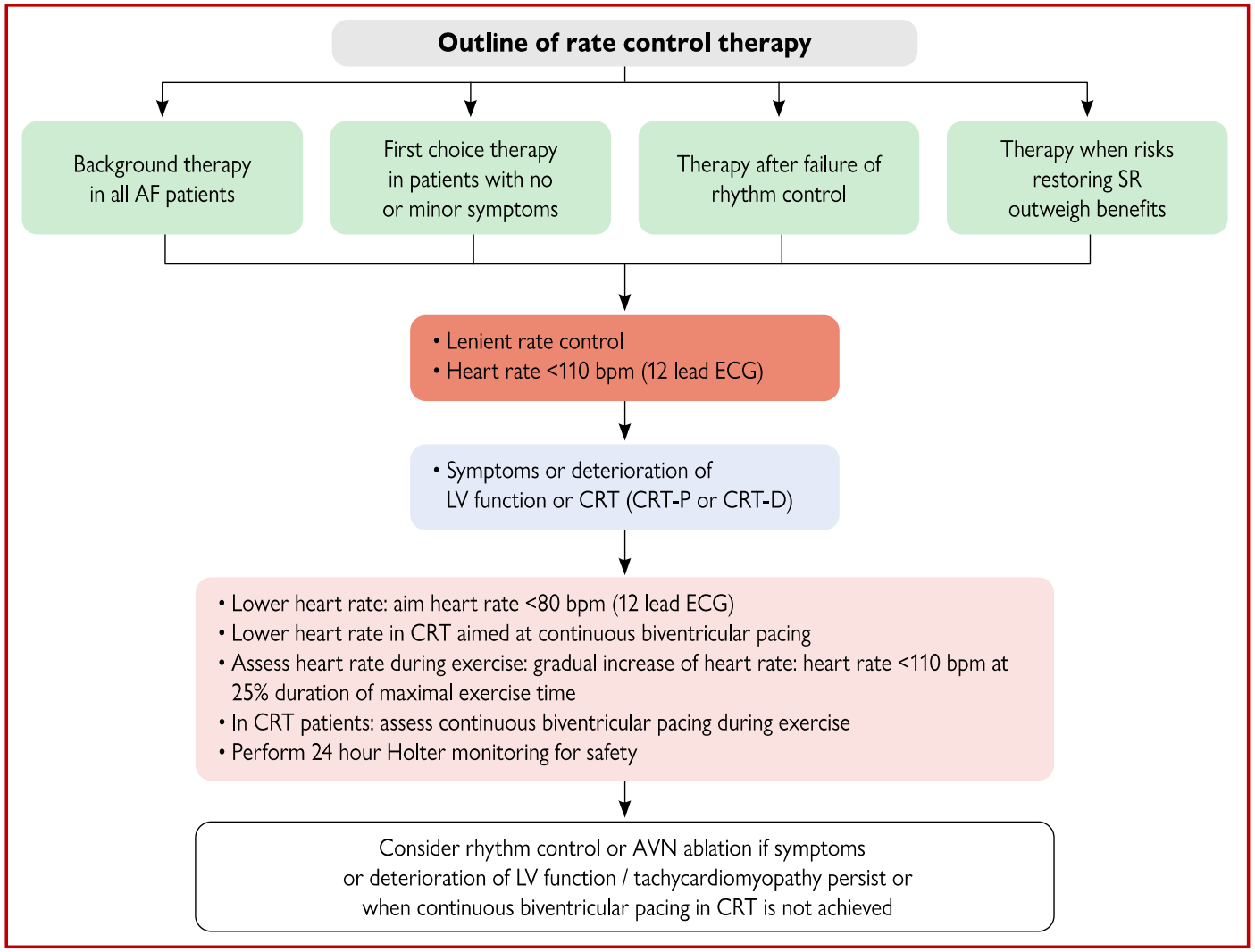
Recommendations for the prevention of thromboembolic events in AF (5)

Recommendations for occlusion or exclusion of the LAA	Class	Level
LAA occlusion may be considered for stroke prevention in patients with AF and contraindications for long-term anticoagulant treatment (e.g. intracranial bleeding without a reversible cause).	IIb	B
Surgical occlusion or exclusion of the LAA may be considered for stroke prevention in patients with AF undergoing cardiac surgery.	IIb	C

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Figure 13 Outline of rate control therapy



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Table 13 Drugs for rate control in AF[a] (1)

	Intravenous administration	Usual oral maintenance dose	Contraindicated
Beta-blockers[b]			
Metoprolol tartrate	2.5–5 mg i.v. bolus; up to 4 doses	25–100 mg *b.i.d.*	In case of asthma use beta-1-blockers Contraindicated in acute HF and history of severe bronchospasm
Metoprolol XL (succinate)	N/A	50–400 mg *o.d.*	
Bisoprolol	N/A	1.25–20 mg *o.d.*	
Atenolol[c]	N/A	25–100 mg *o.d.*	
Esmolol	500 µg/kg i.v. bolus over 1 min, followed by 50–300 µg/kg/min	N/A	
Landiolol	100 µg/kg i.v. bolus over 1 min, followed by 10–40 µg/kg/min; in patients with cardiac dysfunction: 1-10 µg/kg/min	N/A	
Nebivolol	N/A	2.5–10 mg *o.d.*	
Carvedilol	N/A	3.125–50 mg *b.i.d.*	

[a]All rate control drugs are contraindicated in Wolff–Parkinson–White syndrome, also i.v. amiodarone. [b]Other beta-blockers are available but not recommended as specific rate control therapy in AF and therefore not mentioned here (e.g. propranolol and labetalol). [c]No data on atenolol; should not be used in HFrEF.

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Table 13 Drugs for rate control in AF[a] (2)

	Intravenous administration	Usual oral maintenance dose	Contraindicated
Non-dihydropyridine calcium channel antagonists			
Verapamil	2.5–10 mg i.v. bolus over 5 min	40 mg b.i.d. to 480 mg (extended release) o.d.	Contraindicated in HFrEF Adapt doses in hepatic and renal impairment
Diltiazem	0.25 mg/kg i.v. bolus over 5 min, then 5–15 mg/h	60 mg t.i.d. to 360 mg (extended release) o.d.	
Digitalis glycosides			
Digoxin	0.5 mg i.v. bolus (0.75–1.5 mg over 24 hours in divided doses)	0.0625–0.25 mg o.d.	High plasma levels associated with increased mortality Check renal function before starting and adapt dose in CKD patients
Digitoxin	0.4–0.6 mg	0.05–0.1 mg o.d.	High plasma levels associated with increased mortality

[a]All rate control drugs are contraindicated in Wolff–Parkinson–White syndrome, also i.v. amiodarone.

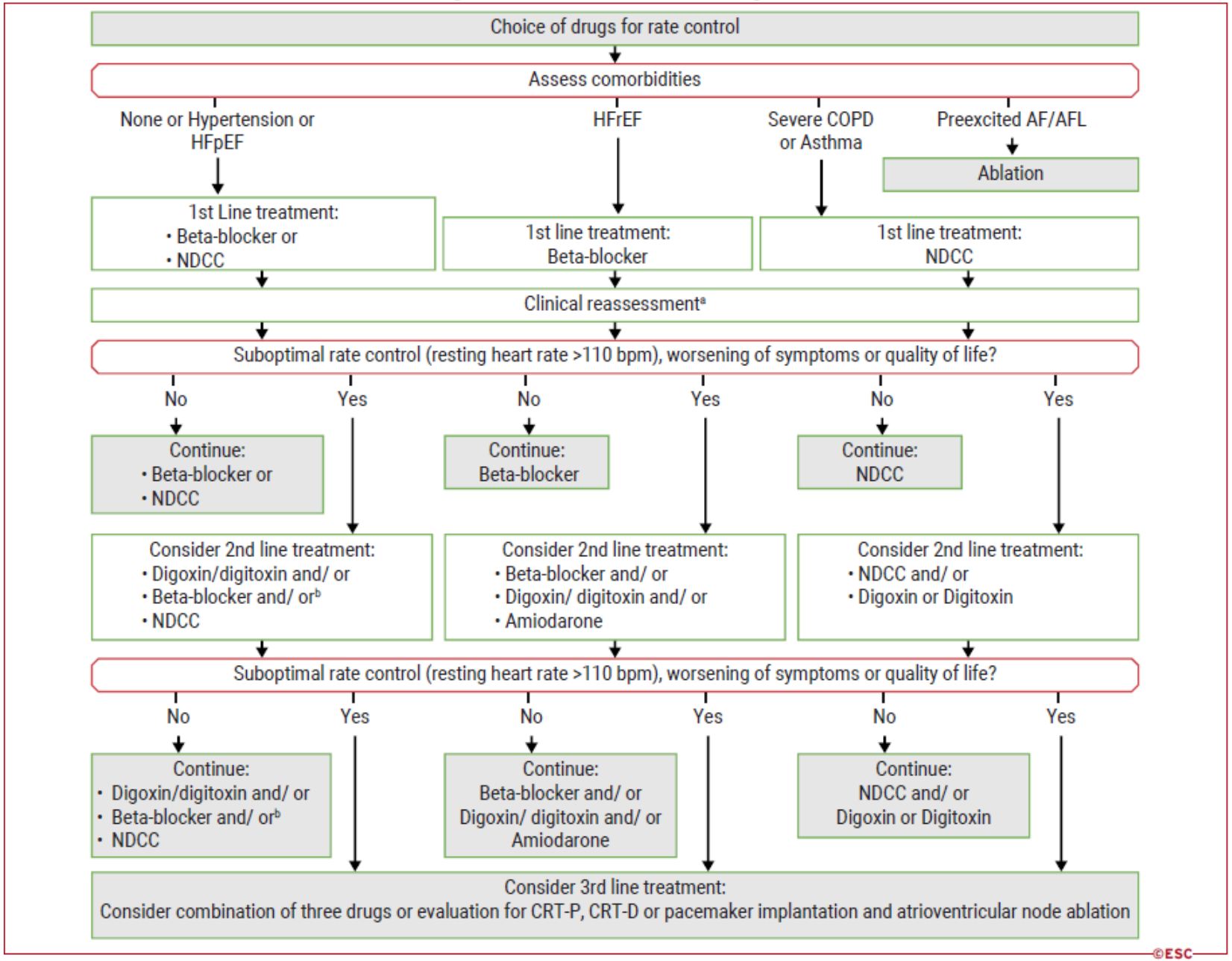
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Table 13 Drugs for rate control in AF[a] (3)

	Intravenous administration	Usual oral maintenance dose	Contraindicated
Other			
Amiodarone	300 mg i.v. diluted in 250 mL 5% dextrose over 30–60 min (preferably via central venous cannula), followed by 900–1200 mg i.v. over 24 hours diluted in 500–1000 mL via a central venous cannula	200 mg o.d. after loading 3×200 mg daily over 4 weeks, then 200 mg daily[d](reduce other rate controlling drugs according to heart rate)	In case of thyroid disease, only if no other options

[a]All rate control drugs are contraindicated in Wolff–Parkinson–White syndrome, also i.v. amiodarone.
[d]Loading regimen may vary; i.v. dosage should be considered when calculating total load.

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[a]Clinical reassessment should be focused on evaluation of resting heart rate, AF/AFL-related symptoms & quality of life. In case suboptimal rate control (resting heart rate >110 bpm), worsening of symptoms or quality of life consider 2nd line &, if necessary, 3rd line treatment options. [b]Careful institution of beta-blocker and NDCC, 24-hour Holter to check for bradycardia.

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Recommendations for ventricular rate control in patients with AF (1)

Recommendations	Class	Level
Beta-blockers, diltiazem, or verapamil are recommended as first-choice drugs to control heart rate in AF patients with LVEF ≥40%.	I	B
Beta-blockers and/or digoxin are recommended to control heart rate in AF patients with LVEF <40%.	I	B
Combination therapy comprising different rate controlling drugs[a] should be considered if a single drug does not achieve the target heart rate.	IIa	B
A resting heart rate of <110 bpm (i.e. lenient rate control) should be considered as the initial heart rate target for rate control therapy.	IIa	B

[a]Combining beta-blocker with verapamil or diltiazem should be performed with careful monitoring of heart rate by 24-h ECG to check for bradycardia.

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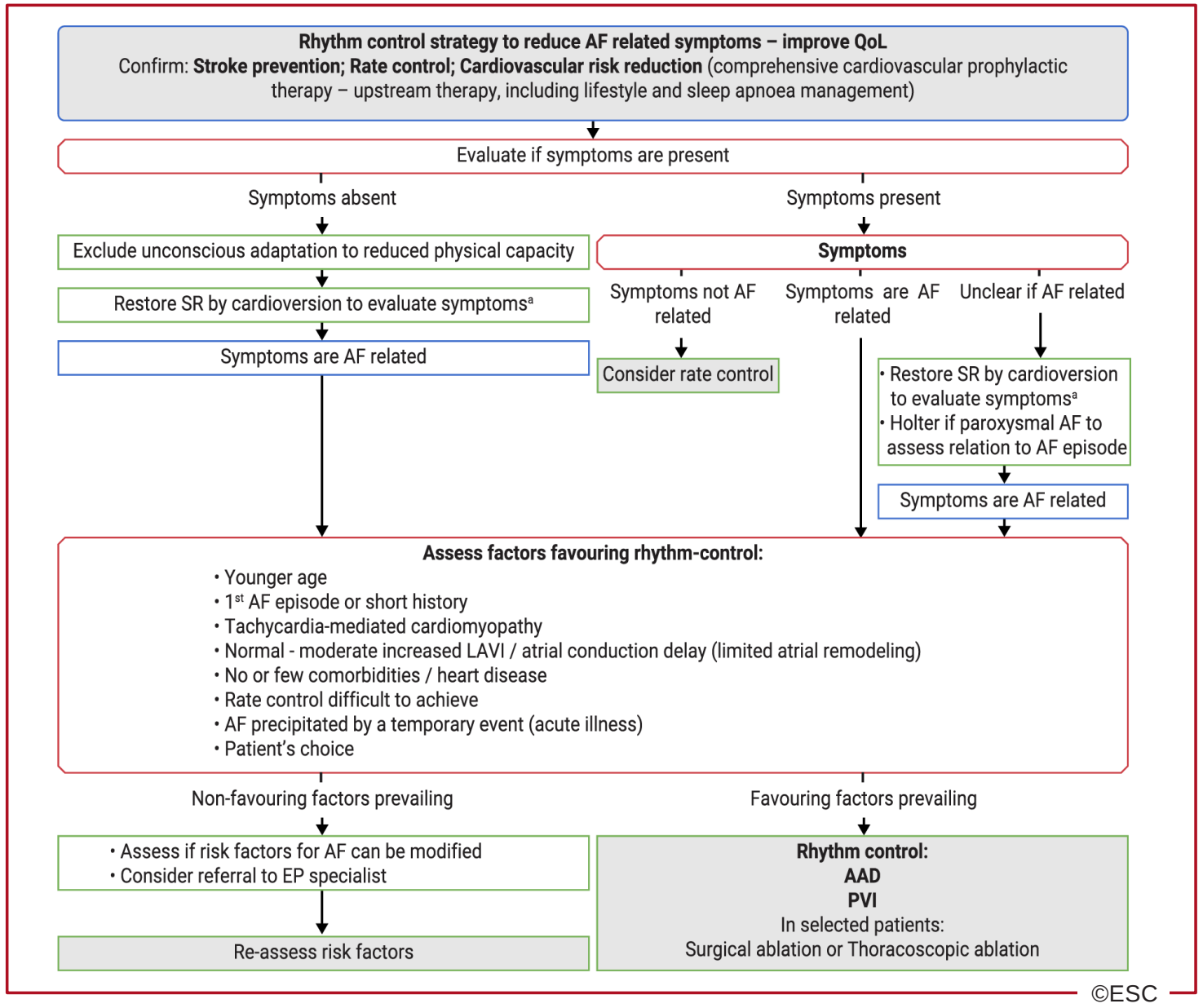
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Recommendations for ventricular rate control in patients with AF (2)

Recommendations	Class	Level
Atrioventricular node ablation should be considered to control heart rate in patients unresponsive or intolerant to intensive rate and rhythm control therapy, and not eligible for rhythm control by LA ablation, accepting that these patients will become pacemaker dependent.	IIa	B
In patients with haemodynamic instability or severely depressed LVEF, intravenous amiodarone may be considered for acute control of heart rate.	IIb	B

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[a]Consider cardioversion to confirm that the absence of symptoms is not due to unconscious adaptation to reduced physical and/or mental capacity.

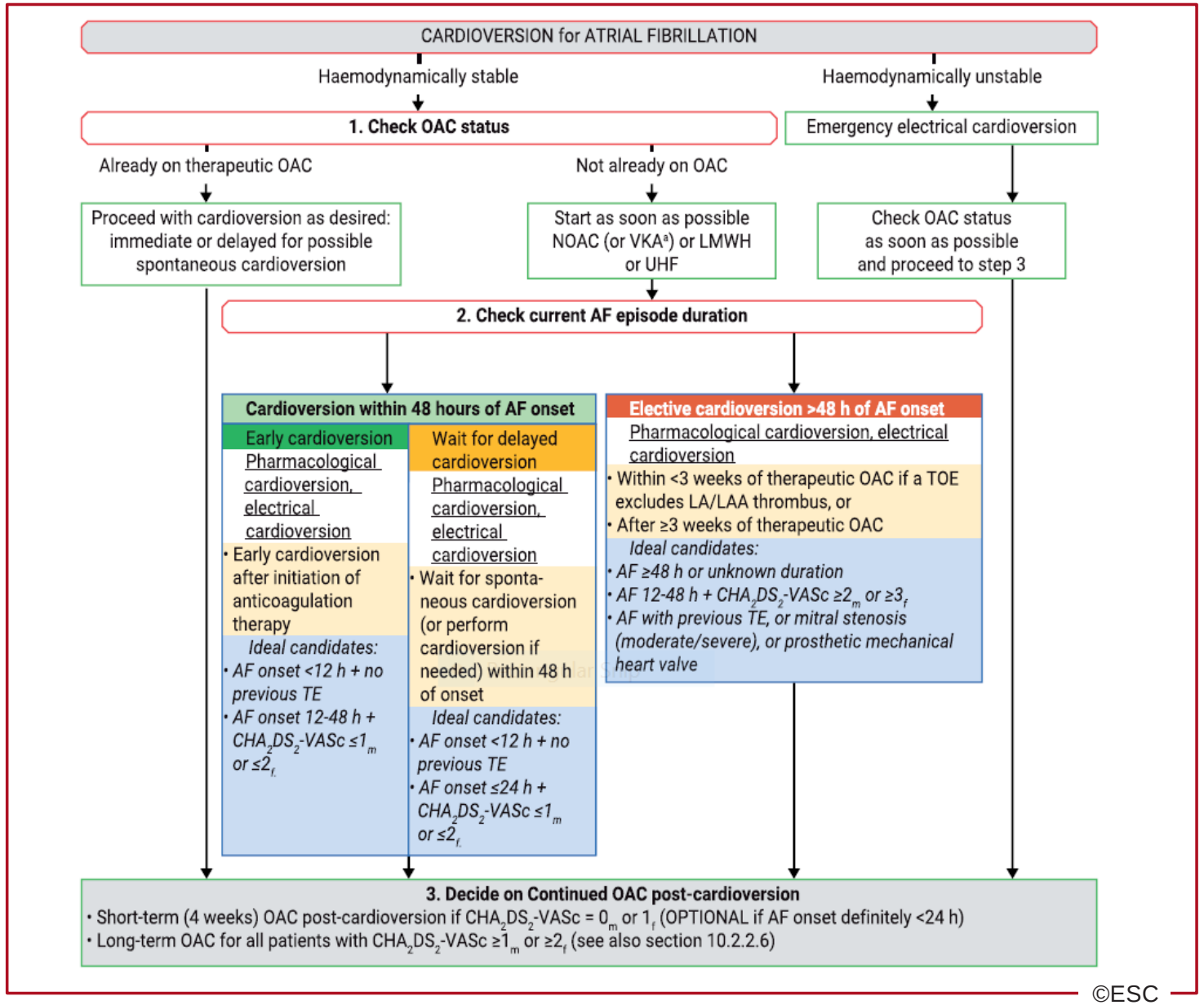
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Recommendations for rhythm control

Recommendations	Class	Level
Rhythm control therapy is recommended for symptom and QoL improvement in symptomatic patients with AF.	I	A

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[a]Alternatively a VKA can be used, accounting for the time needed to achieve therapeutic anticoagulant effect.

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Table 14 Antiarrhythmic drugs used for restoration of sinus rhythm (1)

Antiarrhythmic drugs for restoration of sinus rhythm (pharmacological cardioversion)					
Drug	**Administration route**	**Initial dose For cardioversion**	**Further dosing For cardioversion**	**Acute success rate and expected time to sinus rhythm**	**Contraindications/ Precautions/ comments**
Flecainide[a]	Oral[b] i.v.	200–300 mg 2 mg/kg over 10 min	-	Overall: 59–78% (51% at 3 h, 72% at 8 h)	• Should not be used in ischaemic heart disease and/or significant structural heart disease • May induce hypotension, AFL with 1:1 conduction (in 3.5–5.0% of patients) • Flecainide may induce mild QRS complex widening • Do NOT use for pharmacological cardioversion of AFL
Propafenone[a]	Oral[b] i.v.	450–600 mg 1.5–2 mg/kg over 10 min	-	Oral: 45–55% at 3 h, 69–78% at 8 h; i.v.: 43–89% Up to 6 h	

[a]Most frequently used for cardioversion of AF, available in most countries. [b]May be self-administered by selected outpatients as a 'pill-in-the-pocket' treatment strategy. For more details regarding pharmacokinetic or pharmacodynamic properties refer to EHRA AADs–clinical use and clinical decision making: a consensus document.

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Table 14 Antiarrhythmic drugs used for restoration of sinus rhythm (2)

Antiarrhythmic drugs for restoration of sinus rhythm (pharmacological cardioversion)					
Drug	**Administration route**	**Initial dose For cardioversion**	**Further dosing For cardioversion**	**Acute success rate and expected time to sinus rhythm**	**Contraindications/ Precautions/ comments**
Vernakalant[c]	i.v.	3 mg/kg over 10 min	2 mg/kg over 10 min (10–15 min after the initial dose)	<1 h (50% conversion within 10 min)	• Should not be used in patients with arterial hypotension (SBP <100 mmHg), recent ACS (within 1 month), NYHA III or IV HF, prolonged QT, or severe aortic stenosis • May cause arterial hypotension, QT prolongation, QRS widening, or non-sustained ventricular tachycardia

[c]Not available in some countries. For more details regarding pharmacokinetic or pharmacodynamic properties refer to EHRA AADs–clinical use and clinical decision making: a consensus document.

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Table 14 Antiarrhythmic drugs used for restoration of sinus rhythm (3)

Antiarrhythmic drugs for restoration of sinus rhythm (pharmacological cardioversion)					
Drug	**Administration route**	**Initial dose For cardioversion**	**Further dosing For cardioversion**	**Acute success rate and expected time to sinus rhythm**	**Contraindications/ Precautions/ comments**
Amiodarone[a]	i.v.	5–7 mg/kg over 1–2 h	50 mg/h (maximum 1.2 g for 24 h)	44% 8–12 h to several days	• May cause phlebitis (use a large peripheral vein, avoid i.v. administration >24 hours and use preferably volumetric pump) • May cause hypotension, bradycardia/atrioventricular block, QT prolongation • Only if no other options in patients with hyperthyroidism (risk of thyrotoxicosis)

[a]Most frequently used for cardioversion of AF, available in most countries. For more details regarding pharmacokinetic or pharmacodynamic properties refer to EHRA AADs–clinical use and clinical decision making: a consensus document.

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Table 14 Antiarrhythmic drugs used for restoration of sinus rhythm (4)

Antiarrhythmic drugs for restoration of sinus rhythm (pharmacological cardioversion)					
Drug	**Administration route**	**Initial dose For cardioversion**	**Further dosing For cardioversion**	**Acute success rate and expected time to sinus rhythm**	**Contraindications/ Precautions/ comments**
Ibutilide[c]	i.v.	1 mg over 10 min 0.01 mg/kg if body weight <60 kg	1 mg over 10 min (10–20 min after the initial dose)	31–51% (AF) 63–73% (AFL) ≈1 h	• Effective for conversion of AFL • Should not be used in patients with prolonged QT, severe LVH, or low LVEF • Should be used in the setting of a cardiac care unit as it may cause QT prolongation, polymorphic ventricular tachycardia (torsades de pointes) • ECG monitoring for at least 4 hours after administration to detect a proarrhythmic event

[c]Not available in some countries.

For more details regarding pharmacokinetic or pharmacodynamic properties refer to EHRA AADs–clinical use and clinical decision making: a consensus document.

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Table 15 Goals of follow-up after cardioversion of AF

Goals
Early recognition of AF recurrence by ECG recording after cardioversion
Evaluation of the efficacy of rhythm control by symptom assessment
Monitoring of risk for proarrhythmia by regular control of PR, QRS, and QTc intervals
Evaluation of balance between symptoms and side-effects of therapy considering QoL and symptoms
Evaluation of AF-related morbidities and AAD-related side-effects on concomitant cardiovascular conditions and LV function
Optimization of conditions for maintenance of sinus rhythm including cardiovascular risk management (BP control, HF treatment, increasing cardiorespiratory fitness, and other measures.

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Recommendations for cardioversion (1)

Recommendations	Class	Level
For pharmacological cardioversion of new-onset AF, i.v. vernakalant (excluding patients with recent ACS or severe HF) or flecainide or propafenone (excluding patients with severe structural heart disease) is recommended.	I	A
Intravenous amiodarone is recommended for cardioversion of AF in patients with HF or structural heart disease, if delayed cardioversion is consistent with clinical situation.	I	A
Cardioversion of AF (either electrical or pharmacological) is recommended in symptomatic patients with persistent AF as part of rhythm control therapy.	I	B
Pharmacological cardioversion of AF is indicated only in a haemodynamically stable patient, after consideration of the thromboembolic risk.	I	B

Note: For cardioversion in various specific conditions and AF populations see *section 11*.

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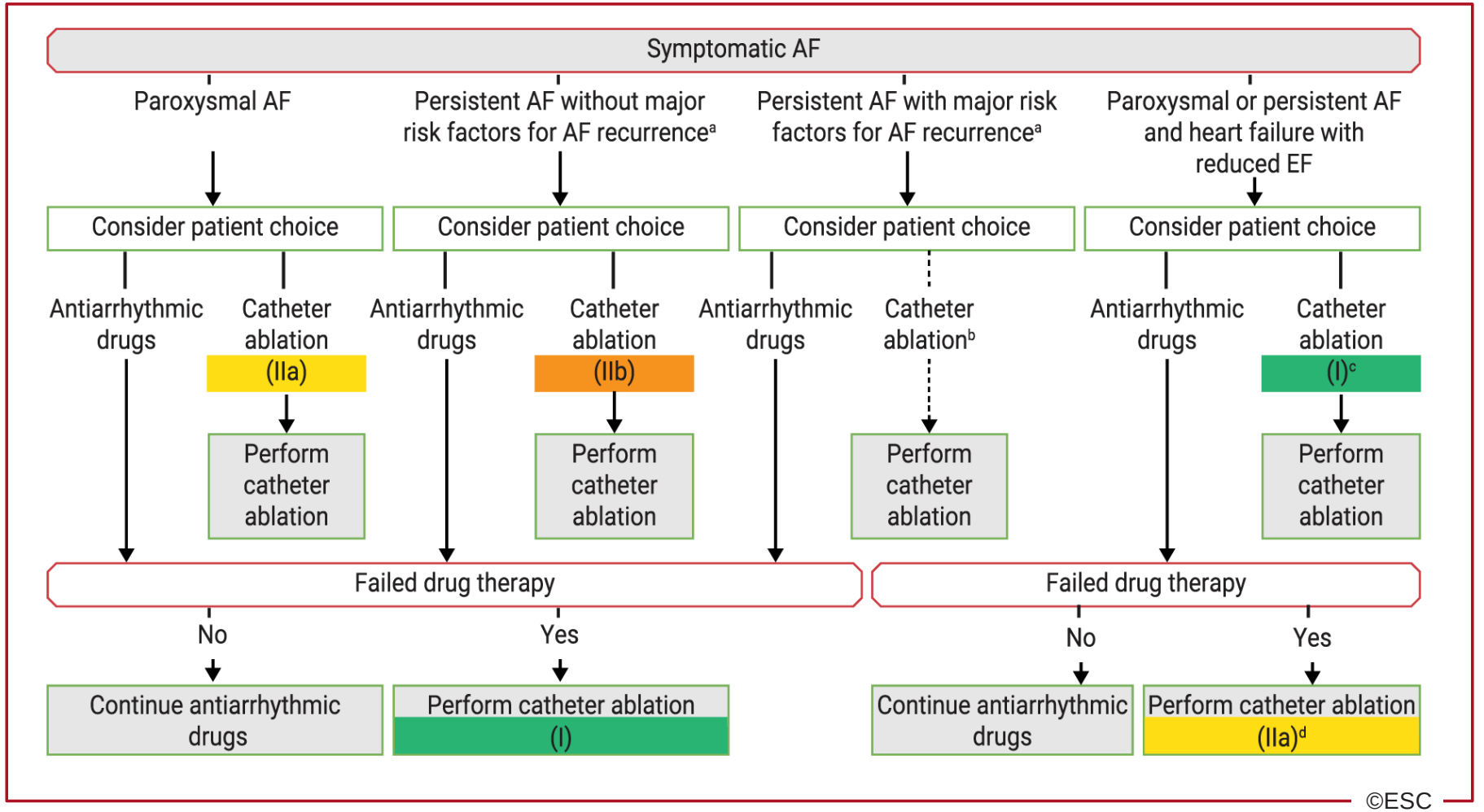
Recommendations for cardioversion (2)

Recommendations	Class	Level
Pretreatment with amiodarone, flecainide, ibutilide, or propafenone should be considered to facilitate the success of electrical cardioversion.	IIa	B
In selected patients with infrequent and recent-onset AF and no significant structural or ischaemic heart disease, a single self-administered oral dose of flecainide or propafenone (‘pill in the pocket’ approach) should be considered for patient-led cardioversion, but only following efficacy and safety assessment.	IIa	B
For patients with sick-sinus syndrome, atrioventricular conduction disturbances or prolonged QTc (>500 ms), pharmacological cardioversion should not be attempted unless risks for proarrhythmia and bradycardia have been considered.	III	C

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Figure 17 Indications for catheter ablation of symptomatic AF



[a]Significantly enlarged LA volume, advanced age, long AF duration, renal dysfunction, and other cardiovascular risk factors. [b]In rare individual circumstances, catheter ablation may be carefully considered as first-line therapy. [c]Recommended to reverse LV dysfunction when tachycardiomyopathy is highly probable.[d]To improve survival and reduce hospitalization.

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Table 16 Procedure-related complications in catheter ablation and thoracoscopic ablation of AF (1)

Complication severity	Complication type	Complication rate	
		Catheter ablation	Thoracoscopic ablation
Life-threatening complications	Periprocedural death	<0.1%	<0.1%
	Oesophageal perforation/fistula	<0.5%	N/A
	Periprocedural thromboembolic event	<1.0%	<1.5%
	Cardiac tamponade	≈1%	<1.0%
Severe complications	Pulmonary vein stenosis	<1.0%	N/A
	Persistent phrenic nerve palsy	<1.0%	N/A
	Vascular complications	2-4%	N/A
	Conversion to sternotomy	N/A	<1.7%
	Pneumothorax	N/A	<6.5%

NA = not available.

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Table 16 Procedure-related complications in catheter ablation and thoracoscopic ablation of AF (2)

Complication severity	Complication type	Complication rate	
		Catheter ablation	Thoracoscopic ablation
Moderate or minor complications	Various	1–2%	1–3%
Complications of unknown significance	Asymptomatic cerebral embolism	5–15%	N/A

NA = not available.

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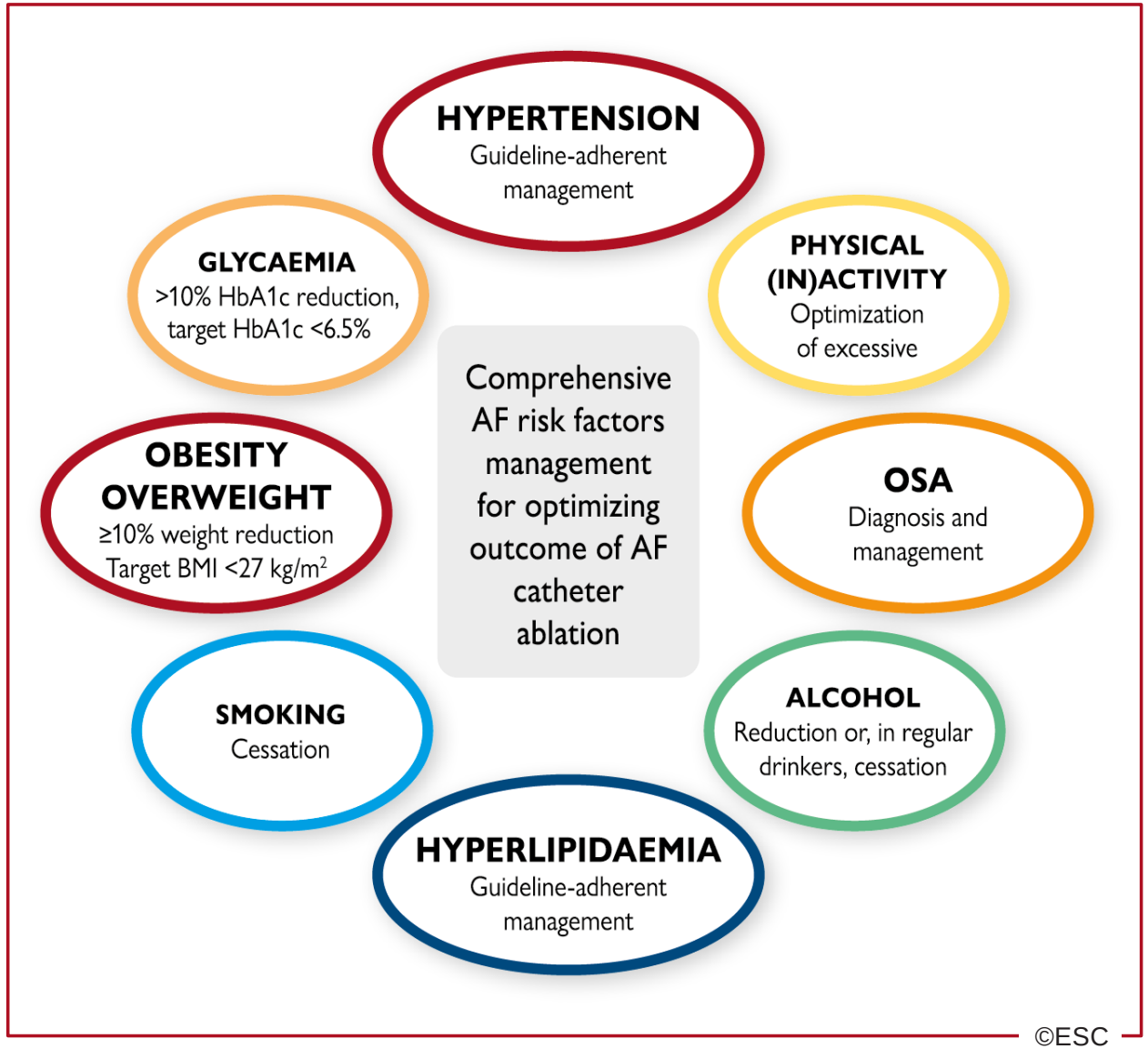


Figure 18 Contribution of AF risk factors to the development of an abnormal substrate translating into poorer outcomes with rhythm control strategies

Several AF risk factors may contribute to the development of LA substrates and thus affect the outcome of AF catheter ablation, predisposing to a higher recurrence rate. Aggressive control of modifiable risk factors may reduce recurrence rate

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Table 17 Key issues in follow-up after AF catheter ablation (1)

Key issues
Recognition and management of complications • Patients must be fully informed about the clinical signs and symptoms of rare but potentially dangerous ablation-related complications that may occur after hospital discharge (e.g. atrio-oesophageal fistula, pulmonary vein stenosis).
Follow-up monitoring: • Useful to assess procedural success and correlate symptom status with rhythm. Recurrences beyond the first month post-ablation are generally predictive of late recurrences, but recurrent symptoms may be due to ectopic beats or other non-sustained arrhythmia; conversely the presence of asymptomatic AF after ablation is well described. • Monitoring may be performed with intermittent ECG, Holter, Patch recordings, external or implanted loop recorder, or smart phone monitor (although the latter has not been validated for such use). Patients should be first reviewed at a minimum of 3 months and annually thereafter.

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Table 17 Key issues in follow-up after AF catheter ablation (2)

Key issues
Management of antiarrhythmic medication and treatment of AF recurrences a. Continuing AAD treatment for 6 weeks to 3 months may reduce early AF recurrences, rehospitalizations and cardioversions during this period. Clinical practice regarding routine AAD treatment after ablation varies and there is no convincing evidence that such treatment is routinely needed. b. Subsequently, AADs may be weaned, ceased, or continued according to symptoms and rhythm status. Recent findings suggest that in AAD-treated patients remaining free of AF at the end of the blanking period, AAD continuation beyond the blanking period reduces arrhythmia recurrences.
Management of anticoagulation therapy a. In general, OAC therapy is continued for 2 months following ablation in all patients. Beyond this time, a decision to continue OAC is determined primarily by the presence of CHA_2DS_2-VASc stroke risk factors rather than the rhythm status (see *section 10.2.2.6*).

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Recommendations for rhythm control/catheter ablation of AF (1)

Recommendations	Class	Level
General recommendations		
For the decision on AF catheter ablation, it is recommended to take into consideration the procedural risks and the major risk factors for AF recurrence following the procedure and discuss them with the patient.	I	B
Repeated PVI procedures should be considered in patients with AF recurrence provided the patient's symptoms were improved after the initial PVI.	IIa	B

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Recommendations for rhythm control/catheter ablation of AF (2)

Recommendations	Class	Level
AF catheter ablation after failure of drug therapy		
AF catheter ablation for PVI is recommended for rhythm control after one failed or intolerant class I or III AAD, to improve symptoms of AF recurrences in patients with	I	
• Paroxysmal AF, or		A
• Persistent AF without major risk factors for AF recurrence, or		A
• Persistent AF with major risk factors for AF recurrence		B
AF catheter ablation for PVI should be considered for rhythm control after one failed or intolerant to beta-blocker treatment to improve symptoms of AF recurrences in patients with paroxysmal and persistent AF.	IIa	B

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Recommendations for rhythm control/catheter ablation of AF (3)

Recommendations	Class	Level
First-line therapy		
AF catheter ablation for PVI should/may be considered as first-line rhythm control therapy to improve symptoms in selected patients with symptomatic:		
• Paroxysmal AF episodes, or	IIa	B
• Persistent AF without major risk factors for AF recurrence.	IIb	C
as an alternative to AAD class I or III, considering patient choice, benefit, and risk.		

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Recommendations for rhythm control/catheter ablation of AF (4)

Recommendations	Class	Level
First-line therapy (continued)		
AF catheter ablation:		
• Is recommended to reverse LV dysfunction in AF patients when tachycardia-induced cardiomyopathy is highly probable, independent of their symptom status.	I	B
• Should be considered in selected AF patients with HF with reduced LVEF to improve survival and reduce HF hospitalization.	IIa	B
AF catheter ablation for PVI should be considered as a strategy to avoid pacemaker implantation in patients with AF-related bradycardia or symptomatic pre-automaticity pause after AF conversion considering the clinical situation.	IIa	C

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Recommendations for rhythm control/catheter ablation of AF (5)

Recommendations	Class	Level
Techniques and technologies		
Complete electrical isolation of the pulmonary veins is recommended during all AF catheter-ablation procedures.	I	A
If patient has history of CTI-dependent AFL or if typical AFL is induced at the time of AF ablation, delivery of a CTI lesion may be considered.	IIb	B
Use of additional ablation lesions beyond PVI (low voltage areas, lines, fragmented activity, ectopic foci, rotors, and others) may be considered but is not well established.	IIb	B

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Recommendations for rhythm control/catheter ablation of AF (6)

Recommendations	Class	Level
Lifestyle modification and other strategies to improve outcomes of ablation		
Weight loss is recommended in obese patients with AF, particularly those who are being evaluated to undergo AF ablation.	I	B
Strict control of risk factors and avoidance of triggers are recommended as part of a rhythm control strategy.	I	B

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Recommendations for surgical ablation of AF

Recommendations	Class	Level
Concomitant AF ablation should be considered in patients undergoing cardiac surgery, balancing the benefits of freedom from atrial arrhythmias and the risk factors for recurrence (left atrial dilatation, years in AF, age, renal dysfunction, and other cardiovascular risk factors).	IIa	A
Thoracoscopic − including hybrid surgical ablation − procedures should be considered in patients who have symptomatic paroxysmal or persistent AF refractory to AAD therapy and have failed percutaneous AF ablation, or with evident risk factors for catheter ablation failure, to maintain long-term sinus rhythm. The decision must be supported by an experienced team of electrophysiologists and surgeons.	IIa	B
Thoracoscopic − including hybrid surgical ablation − procedures may be considered in patients with persistent AF with risk factors for recurrence, who remain symptomatic during AF despite at least one failed AAD and who prefer further rhythm control therapy.	IIb	C

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Recommendations for stroke risk management pericardioversion (1)

Recommendations	Class	Level
In patients with AF undergoing cardioversion, NOACs are recommended with at least similar efficacy and safety as warfarin.	I	A
For cardioversion of AF/AFL, effective anticoagulation is recommended for a minimum of 3 weeks before cardioversion.	I	B
TOE is recommended to exclude cardiac thrombus as an alternative to 3-week preprocedural anticoagulation when early cardioversion is planned.	I	B
In patients at risk of stroke, it is recommended that oral anticoagulant therapy is continued long-term after cardioversion according to the long-term anticoagulation recommendations, irrespective of the method of cardioversion, the apparent maintenance of sinus rhythm, or characterization of AF as a 'first-diagnosed episode".	I	B

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Recommendations for stroke risk management pericardioversion (2)

Recommendations	Class	Level
When thrombus is identified on TOE, effective anticoagulation is recommended for at least 3 weeks before cardioversion of AF.	I	B
It is recommended that the importance of adherence and persistence to NOAC treatment both before and after cardioversion is strongly emphasized to patients.	I	C
Effective anticoagulation should be initiated as soon as possible before every cardioversion of AF or AFL.	IIa	B
Early cardioversion can be performed without TOE in patients with an AF duration of <48 hours.	IIa	B

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Recommendations for stroke risk management pericardioversion (3)

Recommendations	Class	Level
In patients with AF duration of >24 hours undergoing cardioversion, therapeutic anticoagulation should be continued for at least 4 weeks, even after successful cardioversion to sinus rhythm (beyond 4 weeks, the decision about long-term OAC treatment is determined by the presence of stroke risk factors).	IIa	B
When thrombus is identified on TOE, a repeat TOE to ensure thrombus resolution should be considered before cardioversion.	IIa	C
In patients with a definite duration of AF ≤24 hours and a very low stroke risk (CHA_2DS_2-VASc of 0 in men or 1 in women) post-cardioversion anticoagulation for 4 weeks may be omitted.	IIb	C

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Recommendations for stroke risk management peri catheter ablation (1)

Recommendations	Class	Level
In AF patients with stroke risk factors not taking OAC before ablation, it is recommended that preprocedural management of stroke risk includes initiation of anticoagulation and: • Preferably, therapeutic OAC for at least 3 weeks before ablation, or	I	C
• Alternatively, the use of TOE to exclude LA thrombus before ablation.	IIa	C
For patients undergoing AF catheter ablation who have been therapeutically anticoagulated with warfarin, dabigatran, rivaroxaban, apixaban, or edoxaban, performance of the ablation procedure without OAC interruption is recommended.	I	A

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Recommendations for stroke risk management peri catheter ablation (2)

Recommendations	Class	Level
After AF catheter ablation, it is recommended that: • Systemic anticoagulation with warfarin or a NOAC is continued for at least 2 months post ablation, and • Long-term continuation of systemic anticoagulation beyond 2 months post ablation is based on the patient's stroke risk profile and not on the apparent success or failure of the ablation procedure.	I	C

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Recommendations for postoperative anticoagulation after AF surgery

Recommendations	Class	Level
Long-term OAC is recommended in patients after AF surgery and appendage closure, based on the patient's thromboembolic risk assessed with the CHA_2DS_2-VASc score.	I	C

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Table 18 Principles of antiarrhythmic drug therapy

Principles
AAD therapy aims to reduce AF-related symptoms
Efficacy of AADs to maintain sinus rhythm is modest
Clinically successful AAD therapy may reduce rather than eliminate AF recurrences
If one AAD ‘fails’, a clinically acceptable response may be achieved by another drug
Drug-induced proarrhythmia or extracardiac side-effects are frequent
Safety rather than efficacy considerations should primarily guide the choice of AAD

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Table 19 Rules to initiate antiarrhythmic drugs for long-term rhythm control in AF (1)

Consideration	Criteria
Indication for AAD	• **Is the patient symptomatic?** • Are AF symptoms severe enough (EHRA class) to justify AAD use? • Are there associated conditions predicting poor tolerance of AF episodes?
When to start AAD	• **Usually not for the first episode, but it may enhance efficacy of cardioversion**
How to choose among AADs	• **Minimize proarrhythmic risk and organ toxicity** Evaluate for: • basal ECG abnormalities (QRS duration, PR, QTc) and possible interference with AAD • impact on LV function • important pharmacokinetic and pharmacodynamic interactions (i.e. antithrombotic drugs) • **Risk factors for proarrhythmia may be dynamic and change over time**

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Table 19 Rules to initiate antiarrhythmic drugs for long-term rhythm control in AF (2)

Consideration	Criteria
How to minimize proarrhythmic risk	• **Evaluate ECG after the treatment, as indicated in these Guidelines** • Evaluate periodically for organ toxicity (amiodarone) • Long-term Holter monitoring and exercise test in selected cases • Avoid AAD combinations
How to verify efficacy	• **Estimate AF burden under therapy (ask patient for noting episodes)** • If the patient is already on AAD and it was effective but was stopped because of intolerance, choose preferably from the same class
Adjuvant interventions and hybrid therapy	• **In patients with atrioventricular conduction abnormalities and/or sinus node dysfunction, pacemaker implantation should be considered if AAD therapy is deemed necessary** • Short-term AAD therapy could prevent early recurrences after AF ablation

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (1)

Drug	Administration route	Dose	Contraindications/precautions/comments
Amiodarone	Oral	3 × 200 mg daily over 4 weeks, then 200 mg daily	• The most effective AAD • RCTs showed lower AF recurrence compared with sotalol and dronedarone • Also reduces ventricular rate (for 10–12 bpm), safe in patients with HF • Concomitant use with other QT-prolonging drugs with caution • Concomitant use with VKAs or digitalis (their dose should be reduced) • Increased risk of myopathy when used with statins • Requires regular surveillance for liver, lung, and thyroid toxicity

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (2)

Drug	Administration route	Dose	Contraindications/precautions/comments
Amiodarone (continued)	Oral	3 × 200 mg daily over 4 weeks, then 200 mg daily	• Has atrioventricular nodal-slowing properties, but should not be used as first intention for rate control • QT prolongation is common but rarely associated with torsades de pointes (<0.5%) • Torsades de pointes occurs infrequently during treatment with amiodarone (the proarrhythmia caution requires QT-interval and TU-wave monitoring) • Should be discontinued in case of excessive QT prolongation (>500 ms) • ECG at baseline, after 4 weeks

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (3)

Drug	Administration route	Dose	Contraindications/precautions/comments
Amiodarone (continued)	Oral	3 × 200 mg daily over 4 weeks, then 200 mg daily	• Contraindicated in manifest hyperthyroidism • Numerous and frequent extracardiac side-effects may warrant discontinuation of amiodarone, thus making it a second-line treatment when other choices are possible

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (4)

Drug	Administration route	Dose	Contraindications/precautions/comments
Flecainide Flecainide slow release	Oral	100–200 mg b.i.d., or 200 mg once daily (flecainide slow release)	• Effective in preventing recurrence of AF • Should not be used in patients with CrCl <35 mL/min/1.73 m^2 and significant liver disease • Both are contraindicated in patients with ischaemic heart disease or reduced LVEF • Should be discontinued in case of QRS widening >25% above baseline and patients with left bundle-branch block or any other conduction block >120 ms • Caution when sinoatrial/atrioventricular conduction disturbances present • CYP2D6 inhibitors increase concentration[b]

[b]Caution is needed when using any AAD in patients with conduction-system disease (e.g. sinoatrial or atrioventricular node disease).

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (5)

Drug	Administration route	Dose	Contraindications/precautions/comments
Flecainide Flecainide slow release (continued)	Oral	100–200 mg b.i.d., or 200 mg once daily (flecainide slow release)	• May increase AFL cycle length, thus promoting 1:1 atrioventricular conduction and increasing ventricular rate. This risk can be reduced by concomitant administration of an atrioventricular nodal-blocking drug such as a beta-blocker or non-dihydropyridine calcium-channel blocker • In patients properly screened for propensity to proarrhythmias, both flecainide and propafenone are associated with a low proarrhythmic risk • ECG at baseline, after 1–2 weeks

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (6)

Drug	Administration route	Dose	Contraindications/precautions/comments
Propafenone Propafenone slow release	Oral	150–300 mg three times daily, or 225–425 mg b.i.d. (propafenone slow release)	• Should not be used in patients with significant renal or liver disease, ischaemic heart disease, reduced LV systolic function, or asthma • Should be discontinued in case of QRS widening >25% above baseline and in patients left bundle-branch block and any other conduction block >120 ms • Caution when sinoatrial/atrioventricular conduction disturbances present • Increases concentration of warfarin/acenocoumarin and digoxin when used in combination[b]

[b]Caution is needed when using any AAD in patients with conduction-system disease (e.g. sinoatrial or atrioventricular node disease).

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (7)

Drug	Administration route	Dose	Contraindications/precautions/comments
Propafenone Propafenone slow release (continued)	Oral	150–300 mg three times daily, or 225–425 mg b.i.d. (propafenone slow release)	• May increase AFL cycle length, thus promoting 1:1 atrioventricular conduction and increasing ventricular rate • ECG at baseline and after 1–2 weeks

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (8)

Drug	Administration route	Dose	Contraindications/precautions/comments
Dronedarone	Oral	400 mg b.i.d.	• Less effective than amiodarone in rhythm control but has very few extracardiac side-effects • Reduces cardiovascular hospitalizations and death in patients with paroxysmal or persistent AF or AFL and cardiovascular comorbidity • Associated with increased mortality in patients with recent decompensated HF or permanent AF • Dronedarone has the most solid safety data and may thus be a preferable first choice, however not indicated in patients with HF and permanent AF

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (9)

Drug	Administration route	Dose	Contraindications/precautions/comments
Dronedarone (continued)	Oral	400 mg b.i.d.	• Should not be used in NYHA class III or IV or unstable HF, in combination with QT-prolonging drugs or with strong CYP3A4 inhibitors (e.g. verapamil, diltiazem) and in patients with CrCl <30 mL/min • Concomitant use with dabigatran is contraindicated • Combination with digoxin may significantly increase digoxin serum concentration • When used with digitalis or beta-blockers their doses should be reduced • Should be discontinued in case of excessive QT prolongation (>500 ms or >60 ms increase)

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (10)

Drug	Administration route	Dose	Contraindications/precautions/comments
Dronedarone (continued)	Oral	400 mg b.i.d.	• A modest increase in serum creatinine is common and reflects drug-induced reduction in CrCl rather than a decline in renal function • Has atrioventricular nodal-slowing properties • ECG at baseline and after 4 weeks

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (11)

Drug	Administration route	Dose	Contraindications/precautions/comments
Sotalol (d,l racemic mixture)	Oral	80–160 mg b.i.d.	• Only class III effects if dosing >160 mg daily • Considering its safety and efficacy and potential drug alternatives, sotalol should be used with a caution • Should not be used in patients with HFrEF, significant LVH, prolonged QT, asthma, hypokalaemia, or CrCl <30 mL/min • Dose-related torsades de pointes may occur in >2% of patients • Should be discontinued in case of excessive QT prolongation (>500 ms or >60 ms increase)

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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (12)

Drug	Administration route	Dose	Contraindications/precautions/comments
Sotalol (d,l racemic mixture) (continued)	Oral	80–160 mg b.i.d.	• The potassium channel-blocking effect increases with increasing dose and, consequently, the risk of ventricular proarrhythmia (torsades de pointes) increases • Observational data and a recent meta-analysis revealed a correlation with an increased all-cause mortality, whereas a nationwide registry analysis and two RCTs found no evidence for increased safety concerns with sotalol • ECG at baseline, after 1 day and after 1–2 weeks

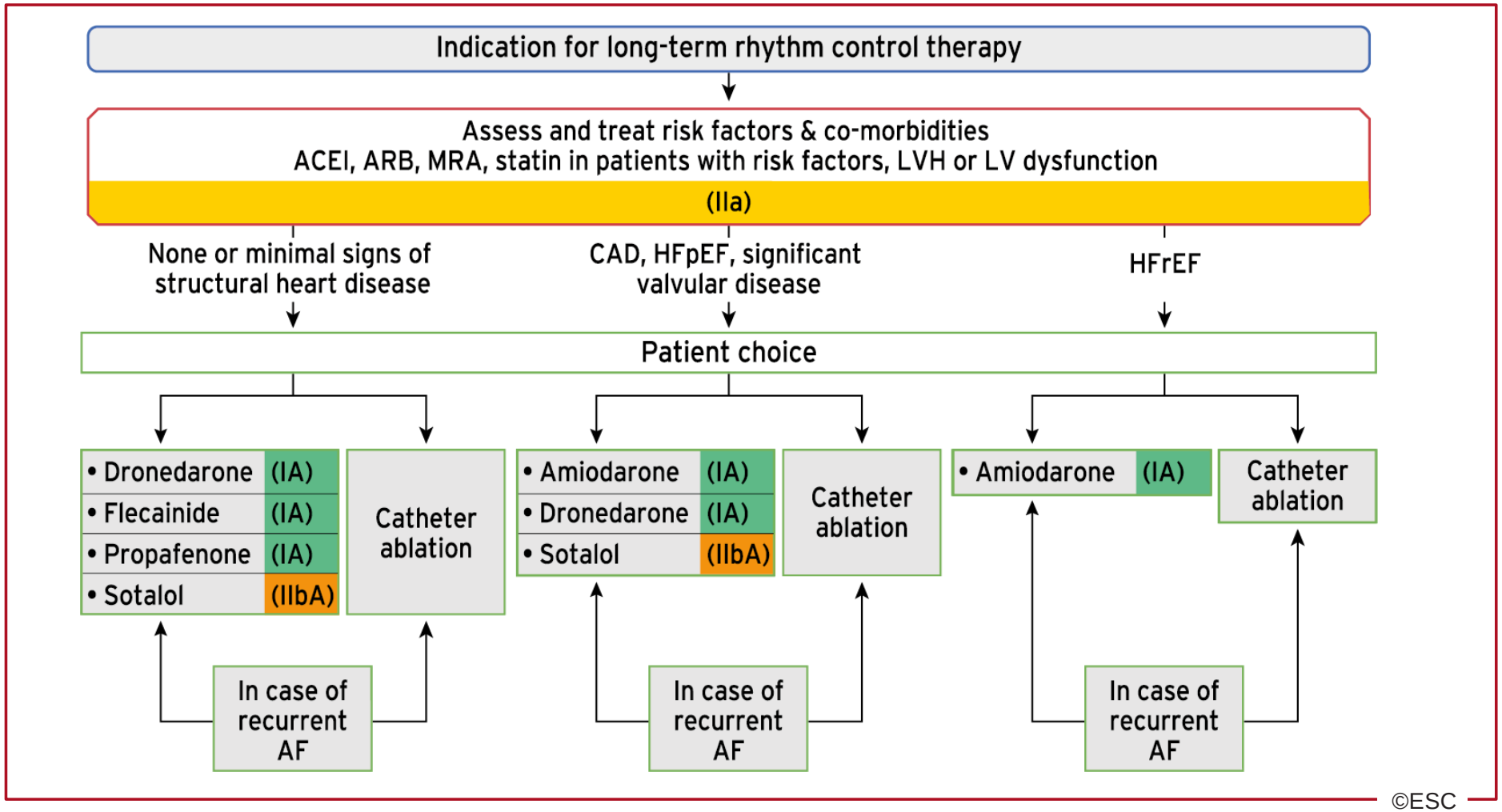
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Table 20 AADs used for long-term maintenance of sinus rhythm in AF patients (13)

Drug	Administration route	Dose	Contraindications/precautions/comments
Disopyramide	Oral	100–400 mg two or t.i.d. (maximum 800 mg/24 h)	• Associated with significantly increased mortality, and rarely used for rhythm control in AF. Should not be used in patients with a structural heart disease. Rarely used for rhythm control in AF patients, due to increased mortality and frequent intolerance to side-effects • May be useful in 'vagal' AF occurring in athletes or during sleep • Reduces LV outflow obstruction and symptoms in patients with HCM

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Figure 19 Long-term rhythm control therapy



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Table 21 Non-antiarrhythmic drugs with antiarrhythmic properties (upstream therapy) (1)

Drugs	Comment
ACEi, ARBs	Activated renin-angiotensin-aldosterone system is up-regulated in AF. ACEi and ARBs showed encouraging results in preventing AF in preclinical studies. As suggested by retrospective analyses and studies where AF was a prespecified secondary endpoint, ACEi/ARBs could prevent new-onset AF in patients with LV dysfunction, LVH, or hypertension. As initial treatment, ACEi and ARBs seem to be superior to other antihypertensive regimens, but ARBs did not reduce AF burden in patients without structural heart disease. Despite several positive small-scale prospective studies and retrospective analyses, larger RCTs have shown controversial results and failed to confirm the role of ACEi or ARBs in secondary (post-cardioversion) prevention of AF. The multifactorial pathways for AF promotion and study design could explain these negative results and should not discourage the use of ACEi or ARB to AAD in patients with structural heart disease.

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Table 21 Non-antiarrhythmic drugs with antiarrhythmic properties (upstream therapy) (2)

Drugs	Comment
MRAs	Aldosterone is implicated in inducibility and perpetuation of AF. Evidence from RCTs showed that MRAs reduced new-onset atrial arrhythmias in patients with HFrEF in parallel with improvement of other cardiovascular outcomes. Recently, the positive impact of MRAs was also shown in patients with HFpEF irrespective of baseline AF status. Regarding other renin-angiotensin-aldosterone system inhibitors, the role of MRAs as upstream therapy in rhythm control strategy for patients with HF and AF has not been clarified. As AF is a marker of HF severity, the beneficial antiarrhythmic effect could be driven indirectly, through improvement of HF. A recent meta-analysis showed that MRAs significantly reduced new-onset AF and recurrent AF, but not postoperative AF.
Beta-blockers	Several small studies suggested a lower AF recurrence rate with beta-blockers, with a comparable efficacy with sotalol. However, most evidence pleads against a significant role of beta-blockers in preventing AF. The observed beneficial effect could also result from transformation of clinically manifested AF to silent AF, because of the rate control with beta-blockers.

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Table 21 Non-antiarrhythmic drugs with antiarrhythmic properties (upstream therapy) (3)

Drugs	Comment
Statins	Statins are attractive candidates for upstream therapy, as the role of inflammation in AF is well established. However, in an adequately designed RCT, statins failed to show a beneficial effect, and their preventive effect was not confirmed in other settings. Specific patient groups in whom statins could induce reverse remodelling are not identified yet, but findings from the CARAF registry suggested that AF patients already on beta-blockers could benefit from statin therapy. Polyunsaturated fatty acids also failed to show convincing benefit in preventing AF.

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Recommendations for long-term antiarrhythmic drugs (1)

Recommendations	Class	Level
Flecainide or propafenone is recommended for long-term rhythm control in AF patients with normal LV function and without structural heart disease, including significant LVH and myocardial ischaemia.	I	A
Dronedarone is recommended for long-term rhythm control in AF patients with: • Normal or mildly impaired (but stable) LV function, or • HFpEF, ischaemic, or VHD.	I	A
Amiodarone is recommended for long-term rhythm control in all AF patients, including those with HFrEF. However, owing to its extracardiac toxicity, other AADs should be considered first whenever possible.	I	A
In AF patients treated with sotalol, close monitoring of QT interval, serum potassium levels, CrCl, and other proarrhythmia risk factors is recommended.	I	B

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Recommendations for long-term antiarrhythmic drugs (2)

Recommendations	Class	Level
In AF patients treated with flecainide for long-term rhythm control, concomitant use of an atrioventricular nodal-blocking drug (if tolerated) should be considered.	IIa	C
Sotalol may be considered for long-term rhythm control in patients with normal LV function or with ischaemic heart disease if close monitoring of QT interval, serum potassium levels, CrCl, and other proarrhythmia risk factors is provided.	IIb	A
AAD therapy is not recommended in patients with permanent AF under rate control and in patients with advanced conduction disturbances unless antibradycardia pacing is provided.	III	C

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Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with AF (1)

ESC European Society of Cardiology

Recommendations	Class	Level
Identification and management of risk factors and concomitant diseases is recommended as an integral part of treatment in AF patients.	I	B
Modification of unhealthy lifestyle and targeted therapy of intercurrent conditions is recommended to reduce AF burden and symptom severity.	I	B
Opportunistic screening for AF is recommended in hypertensive patients.	I	B
Attention to good BP control is recommended in AF patients with hypertension to reduce AF recurrences and risk of stroke and bleeding.	I	B
In obese patients with AF, weight loss together with management of other risk factors should be considered to reduce AF incidence, AF progression, AF recurrences, and symptoms.	IIa	B

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Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with AF (2)

ESC European Society of Cardiology

Recommendations	Class	Level
Advice and management to avoid alcohol excess should be considered for AF prevention and in AF patients considered for OAC therapy	IIa	B
Physical activity should be considered to help prevent AF incidence or recurrence, with the exception of excessive endurance exercise, which may promote AF.	IIa	C
Opportunistic screening for AF should be considered in patients with OSA.	IIa	C
Optimal management of OSA may be considered, to reduce AF incidence, AF progression, AF recurrences, and symptoms.	IIb	C

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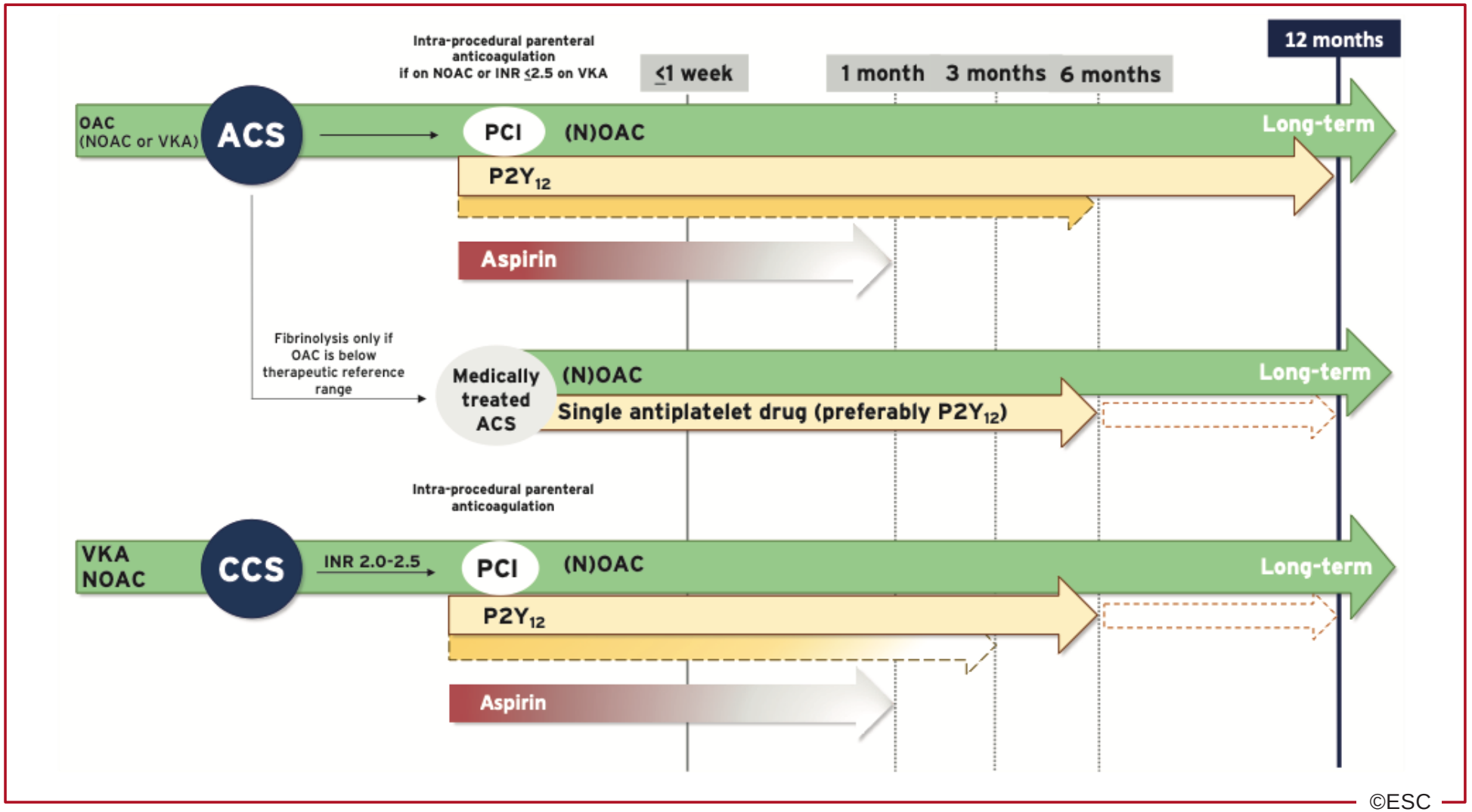
Recommendations for management of AF with haemodynamic instability

Recommendations	Class	Level
Emergency electrical cardioversion is recommended in AF patients with acute or worsening haemodynamic instability.	I	B
In AF patients with haemodynamic instability, amiodarone may be considered for acute control of heart rate.	IIb	B

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Figure 20 (1) Post-procedural management of patients with AF and ACS/PCI (full-outlined arrows represent a default strategy; graded/dashed arrows show treatment modifications depending on individual patient's ischaemic and bleeding risks)



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Figure 20 (2) Post-procedural management of patients with AF and ACS/PCI (full-outlined arrows represent a default strategy; graded/dashed arrows show treatment modifications depending on individual patient's ischaemic and bleeding risks)

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THROMBOTIC RISK FACTORS

- Diabetes mellitus requiring therapy
- Prior ACS/recurrent myocardial infarction
- Multivessel CAD
- Concomitant PAD
- Premature CAD (occurring at age of <45 y) or accelerated CAD (new lesion within 2 years)
- CKD (eGFR <60 mL/min)
- Clinical presentation (ACS)
- Multivessel stenting
- Complex revascularisation (left main stenting, bifurcation lesion stenting, chronic total occlusion intervention, last patent vessel stenting)
- Prior stent thrombosis on antiplatelet treatment
- Procedural factors (stent expansion, residual dissection, stent length, etc.)

BLEEDING RISK FACTORS

- Hypertension
- Abnormal renal or liver function
- Stroke or ICH history
- Bleeding history or bleeding diathesis (e.g., anaemia with haemoglobin <110 g/L)
- Labile INR (if on VKA)
- Elderly (>65 years)
- Drugs (concomitant OAC and antiplatelet therapy, NSAIDs), excessive alcohol consumption

STRATEGIES TO REDUCE BLEEDING ASSOCIATED WITH PCI

- Radial artery access
- PPIs in patients taking DAPT who are at increased risk of bleeding (e.g., the elderly, dyspepsia, gastro-oesophageal reflux disease, Helicobacter pylori infection, chronic alcohol use)
- Non–administration of unfractionated heparin in patients on VKA with INR >2.5
- Pre-treatment with aspirin only, add a $P2Y_{12}$ inhibitor when coronary anatomy is known or if STEMI
- GP IIb/IIIa inhibitors only for bailout or periprocedural complications
- Shorter duration of combined antithrombotic therapy

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Recommendations for patients with AF and an ACS, PCI, or CCS (1)

Recommendations	Class	Level
General recommendations for patients with AF and an indication for concomitant antiplatelet therapy		
In AF patients eligible for NOACs, it is recommended to use a NOAC[a] in preference to a VKA in combination with antiplatelet therapy.	I	A
In patients at high bleeding risk (HAS-BLED ≥3), rivaroxaban 15 mg o.d. should be considered in preference to rivaroxaban 20 mg o.d. for the duration of concomitant single or DAPT, to mitigate bleeding risk.	IIa	B

[a]See summary of product characteristics for reduced doses or contraindications for each NOAC in patients with CKD, body weight <60 kg, age >75–80 years, and/or drug interactions.

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Recommendations for patients with AF and an ACS, PCI, or CCS (2)

Recommendations	Class	Level
General recommendations for patients with AF and an indication for concomitant antiplatelet therapy (continued)		
In patients at high bleeding risk (HAS-BLED ≥3), dabigatran 110 mg b.i.d. should be considered in preference to dabigatran 150 mg b.i.d. for the duration of concomitant single or DAPT, to mitigate bleeding risk.	**IIa**	**B**
In AF patients with an indication for a VKA in combination with antiplatelet therapy, the VKA dosing should be carefully regulated with a target INR of 2.0–2.5 and TTR >70%.	**IIa**	**B**

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Recommendations for patients with AF and an ACS, PCI, or CCS (3)

Recommendations	Class	Level
Recommendations for AF patients with ACS		
In AF patients with ACS undergoing an uncomplicated PCI, early cessation (≤1 week) of aspirin and continuation of dual therapy with an OAC and a $P2Y_{12}$ inhibitor (preferably clopidogrel) for up to 12 months is recommended if the risk of stent thrombosis[b] is low or if concerns about bleeding risk[c] prevail over concerns about risk of stent thrombosis[b], irrespective of the type of stent used.	I	A
Triple therapy with aspirin, clopidogrel, and an OAC[d] for longer than 1 week after an ACS should be considered when risk of stent thrombosis[b] outweighs the bleeding risk,[c] with the total duration (≤1 month) decided according to assessment of these risks, and the treatment plan should be clearly specified at hospital discharge.	IIa	C

[b]Risk of stent thrombosis encompasses: (i) risk of thrombosis occurring and (ii) risk of death should stent thrombosis occur, both of which relate to anatomical, procedural, and clinical characteristics. Risk factors for CCS patients include stenting of left main stem or last remaining patent artery; suboptimal stent deployment; stent length >60 mm; diabetes mellitus; CKD; bifurcation with two stents implanted; treatment of chronic total occlusion; and previous stent thrombosis on adequate antithrombotic therapy. [c]Bleeding risk in AF patients may be assessed using the HAS-BLED score, which draws attention to modifiable bleeding risk factors; those at high risk (score ≥3) can have more frequent or early review and follow-up. Bleeding risk is highly dynamic and does not remain static, and relying on modifiable bleeding risk factors alone is an inferior strategy to evaluate bleeding risk. [d]When dabigatran is used in triple therapy, dabigatran 110 mg b.i.d may be used instead of 150 mg b.i.d, but the evidence is insufficient.

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Recommendations for patients with AF and an ACS, PCI, or CCS (4)

Recommendations	Class	Level
Recommendations in AF patients with a CCS undergoing PCI		
After uncomplicated PCI, early cessation (≤1 week) of aspirin and continuation of dual therapy with OAC for up to 6 months and clopidogrel is recommended if the risk of stent thrombosis[b] is low or if concerns about bleeding risk[c] prevail over concerns about risk of stent thrombosis,[b] irrespective of the type of stent used.	I	A
Triple therapy with aspirin, clopidogrel, and an OAC[d] for longer than 1 week should be considered when risk of stent thrombosis[b] outweighs the bleeding risk,[c] with the total duration (≤1 month) decided according to assessment of these risks, and the treatment plan should be clearly specified at hospital discharge.	IIa	C

[b]Risk of stent thrombosis encompasses: (i) risk of thrombosis occurring and (ii) risk of death should stent thrombosis occur, both of which relate to anatomical, procedural, and clinical characteristics. Risk factors for CCS patients include stenting of left main stem or last remaining patent artery; suboptimal stent deployment; stent length >60 mm; diabetes mellitus; CKD; bifurcation with two stents implanted; treatment of chronic total occlusion; and previous stent thrombosis on adequate antithrombotic therapy.[c] Bleeding risk in AF patients may be assessed using the HAS-BLED, which draws attention to modifiable bleeding risk factors; those at high risk (score ≥3) can have more frequent or early review and follow-up. Bleeding risk is highly dynamic and does not remain static, and relying on modifiable bleeding risk factors alone is an inferior strategy to evaluate bleeding risk.[d]When dabigatran is used in triple therapy, dabigatran 110 mg b.i.d may be used instead of 150 mg b.i.d, but the evidence is insufficient.

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Recommendations for the search for AF in patients with cryptogenic stroke

Recommendations	Class	Level
In patients with acute ischaemic stroke or TIA and without previously known AF, monitoring for AF is recommended using a short-term ECG recording for at least the first 24 hours, followed by continuous ECG monitoring for at least 72 hours whenever possible.	I	B
In selected[a] stroke patients without previously known AF, additional ECG monitoring using long-term non-invasive ECG monitors or insertable cardiac monitors should be considered, to detect AF.	IIa	B

[a]Not all stroke patients would benefit from prolonged ECG monitoring; those deemed at risk of developing AF (e.g. elderly, with cardiovascular risk factors or comorbidities, indices of LA remodelling, high C_2HEST score, etc.) or those with cryptogenic stroke and stroke characteristics suggestive of an embolic stroke should be scheduled for prolonged ECG monitoring.

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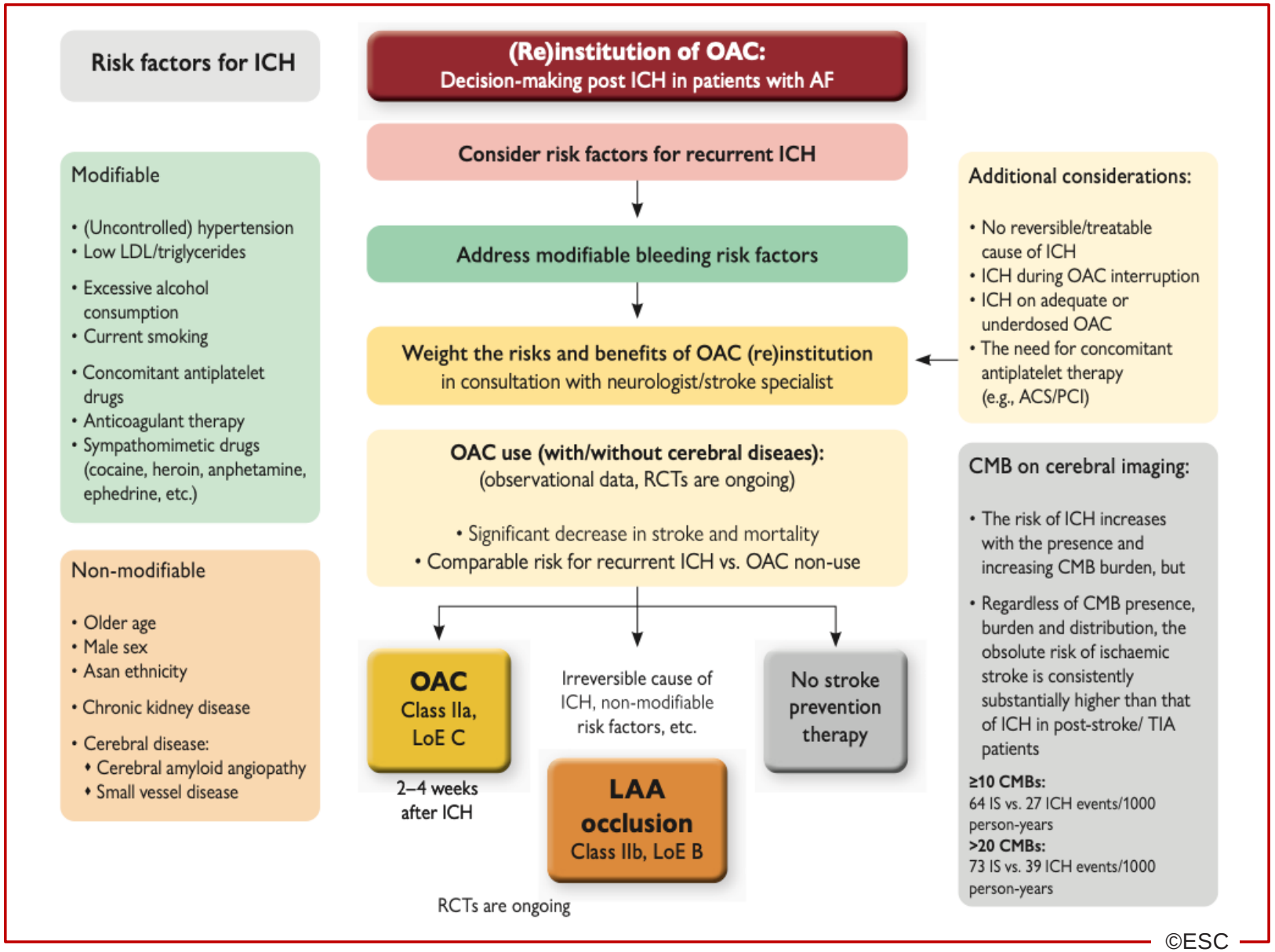


Figure 21
(Re-) initiation of anticoagulation post-intracranial bleeding

A pooled analysis of individual patient data from cohort studies (n=20 322 patients; 38 cohorts; >35 225 patient-years) showed that although cerebral microbleeds can inform regarding the risk for ICH in patients with recent ischaemic stroke/TIA treated with antithrombotic therapy, the absolute risk of ischaemic stroke is substantially higher than that of ICH, regardless of the presence, burden, or location of cerebral microbleeds

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Recommendations for secondary stroke prevention in AF patients after acute ischaemic stroke

Recommendations	Class	Level
In AF patients with an ischaemic stroke or TIA, long-term secondary prevention of stroke using OAC is recommended if there is no strict contraindication to OAC use, with a preference for NOACs over VKAs in NOAC-eligible patients.	I	A
In AF patients presenting with acute ischaemic stroke, very early anticoagulation (<48 hours) using UFH, LMWH, or VKAs is not recommended.	III	B

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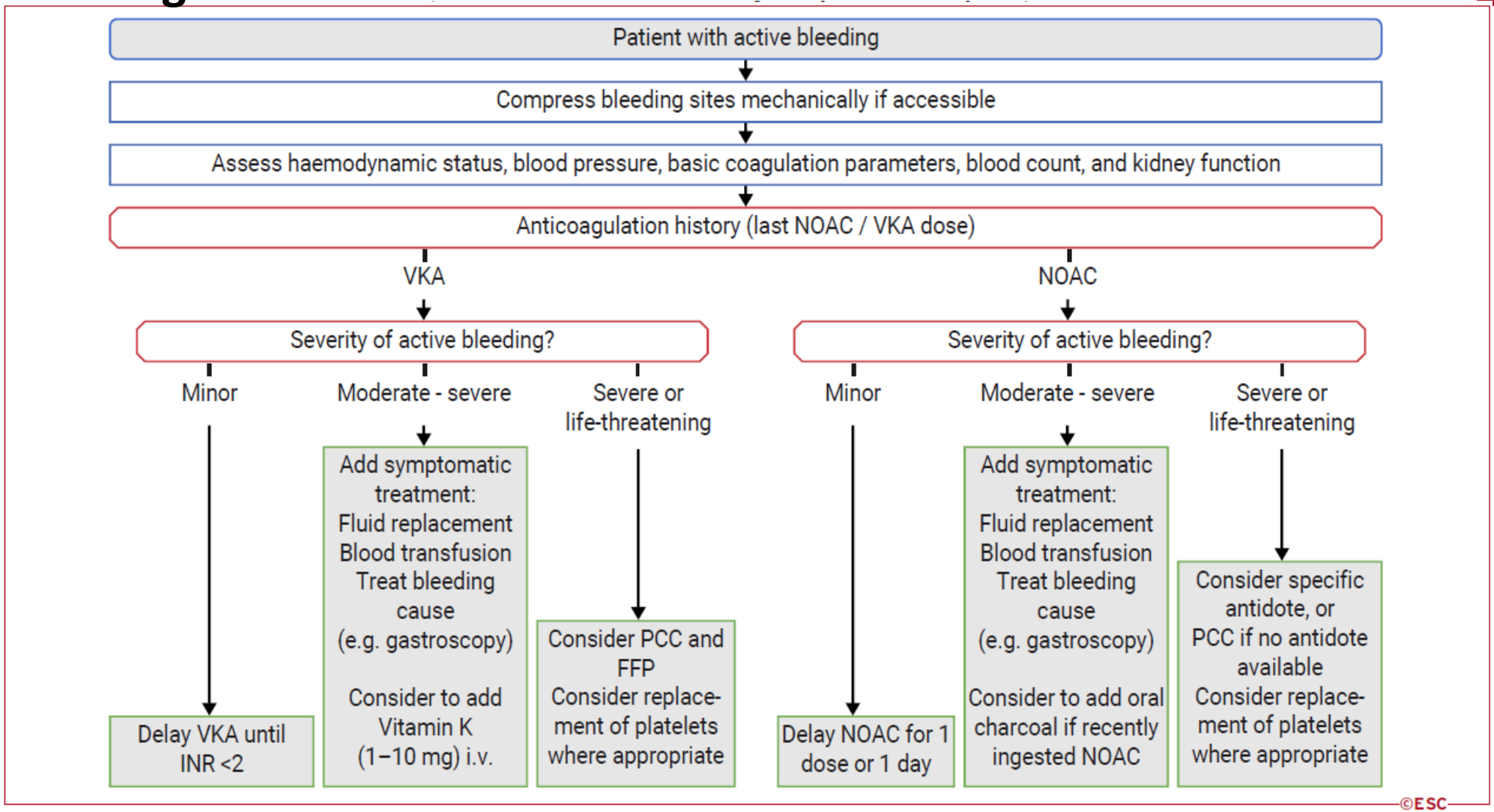
Recommendations for stroke prevention in AF patients after intracranial haemorrhage

Recommendations	Class	Level
In AF patients at high risk of ischaemic stroke, (re-)initiation of OAC, with preference for NOACs over VKAs in NOAC-eligible patients, should be considered in consultation with a neurologist/stroke specialist after: • A trauma-related ICH • Acute spontaneous ICH (which includes subdural, subarachnoid, or intracerebral haemorrhage), after careful consideration of risks and benefits.[a]	IIa	C

[a]A more favourable net benefit is likely with deep ICH or without neuroimaging evidence of cerebral amyloid angiopathy or microbleeds.

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Figure 22 Management of active bleeding in patients receiving anticoagulation



Management of active bleeding in patients receiving anticoagulation (institutions should have an agreed procedure in place).

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Recommendations for the management of active bleeding on OAC

Recommendations	Class	Level
In an AF patient with severe active bleeding, it is recommended to: • Interrupt OAC until the cause of bleeding is identified and active bleeding is resolved; and • Promptly perform specific diagnostic and treatment interventions to identify and manage the cause(s) and source(s) of bleeding.	I	C
Four-factor prothrombin complex concentrates should be considered in AF patients on VKA who develop a severe bleeding complication.	IIa	C

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Recommendations for patients with valvular heart disease and AF

Recommendations	Class	Level
NOACs are contraindicated in patients with a prosthetic mechanical valve.	III	B
Use of NOACs is not recommended in patients with AF and moderate-to-severe mitral stenosis.	III	C

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Recommendations for the management of AF in patients with congenital heart disease (1)

Recommendations	Class	Level
• Oral anticoagulation should be considered in *all* adult patients with intracardiac repair, cyanosis, Fontan palliation, or systemic right ventricle and a history of AF, atrial flutter, or intra-atrial re-entrant tachycardia. • In patients with AF and other congenital heart diseases, anticoagulation should be considered in the presence of one or more non-sex stroke risk factor(s).	IIa	C
Surgery for AF should be considered in patients: • Who need surgical closure of an atrial septal defect and who have a history of symptomatic atrial arrhythmia (atrial ablation should be considered at the time of surgical closure). • Cox maze surgery should be considered in patients with symptomatic AF and an indication for corrective repair of congenital heart defects. The surgery should be done in experienced centres.	IIa	C

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Recommendations for the management of AF in patients with congenital heart disease (2)

Recommendations	Class	Level
AF catheter ablation of atrial arrhythmias associated with congenital heart defects may be considered when performed in experienced centres.	IIb	C
In patients with congenital heart disease, TOE may be considered together with 3-week anticoagulation therapy before cardioversion.	IIb	C

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Recommendations for the management of AF during Pregnancy (1)

Recommendations	Class	Level
Acute management		
Immediate electrical cardioversion[a] is recommended in case of haemodynamic instability or pre-excited AF.	I	C
In pregnant women with HCM, cardioversion[a] should be considered for persistent AF.	IIa	C
Ibutilide or flecainide i.v. may be considered for termination of AF in stable patients with structurally normal hearts.	IIb	C

[a]Cardioversion of AF should generally be preceded by anticoagulation

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Recommendations for the management of AF during Pregnancy (2)

Recommendations	Class	Level
Long-term management (oral administration of drugs)		
Therapeutic anticoagulation with heparin or VKA according to the stage of pregnancy is recommended for patients with AF.	I	C
Beta-selective blockers are recommended for rate control in AF.[b]	I	C
Flecainide[c], propafenone,[c] or sotalol[d] should be considered to prevent AF if atrioventricular nodal-blocking drugs[d] fail.	IIa	C
Digoxin[e] or verapamil[e] should be considered for rate control if beta-blockers fail.	IIa	C

[b]Atenolol have been associated with higher rates of foetal growth retardation and is not recommended.[c]Flecainide and propafenone should be combined with atrioventricular nodal-blocking drugs, but structural heart disease, reduced LV function, and bundle branch block should be excluded. [d]Class III drugs should not be used in prolonged QTc. [e]Atrioventricular nodal-blocking drugs should not be used in patients with pre-excitation on resting ECG or pre-excited AF.
Note that the former A to X categories of drugs – the classification system for counselling of pregnant women requiring drug therapy – was replaced by the Pregnancy and Lactation Labelling Rule, which provides a descriptive risk summary and detailed information on animal and clinical data, by the US FDA in June 2015.

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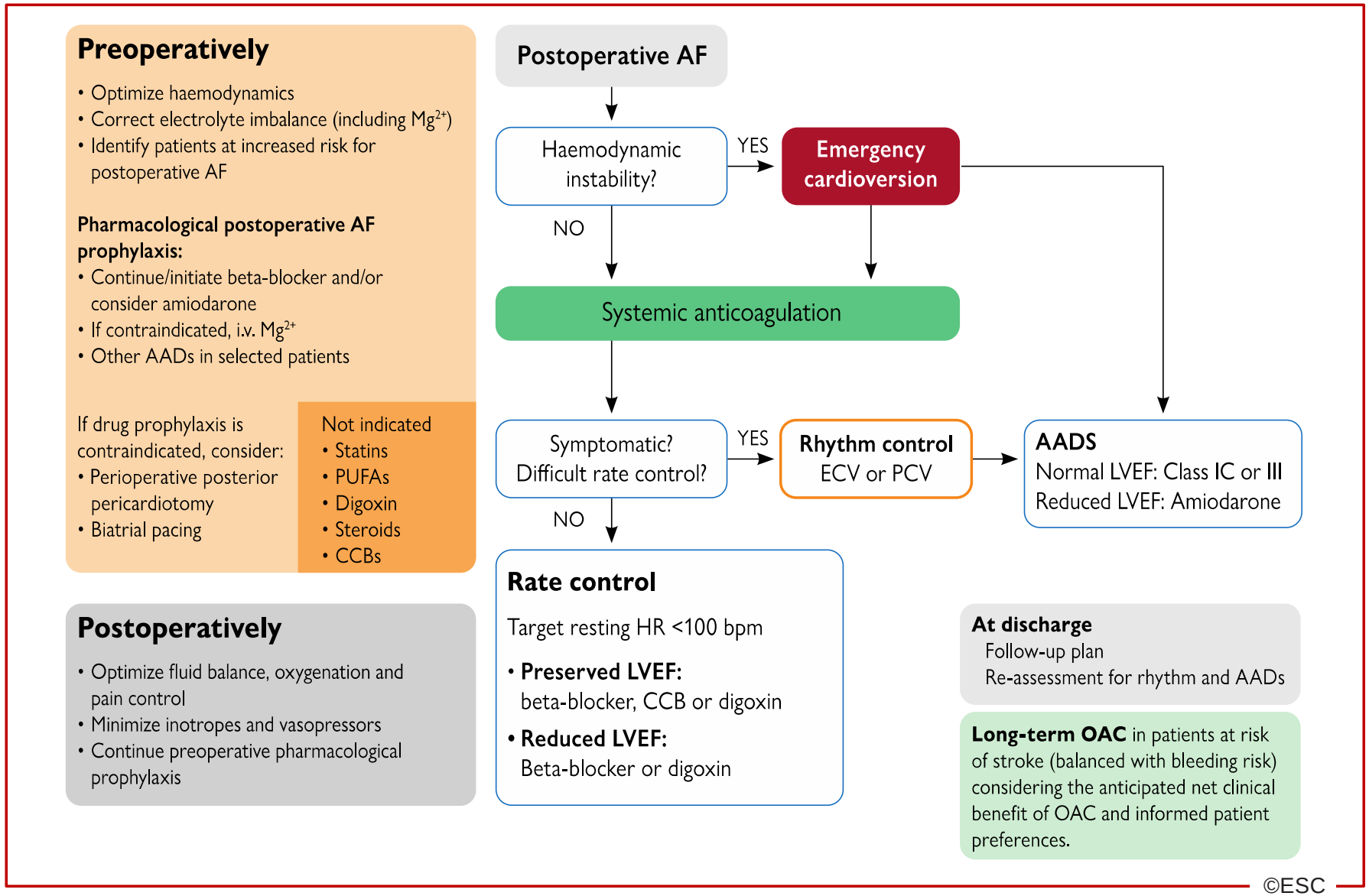
Recommendations for sports activity in patients with AF

Recommendations	Class	Level
It is recommended to counsel professional athletes that long-lasting intense sports participation may promote AF, while moderate physical activity is recommended to prevent AF.	I	B

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Figure 23 Management of postoperative AF



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Recommendations for postoperative AF

Recommendations	Class	Level
Perioperative amiodarone or beta-blocker therapy is recommended for the prevention of postoperative AF after cardiac surgery.	I	A
Long-term OAC therapy to prevent thromboembolic events should be considered in patients at risk for stroke with postoperative AF after non-cardiac surgery, considering the anticipated net clinical benefit of OAC therapy and informed patient preferences.	IIa	B
Long-term OAC therapy to prevent thromboembolic events may be considered in patients at risk for stroke with postoperative AF after cardiac surgery, considering the anticipated net clinical benefit of OAC therapy and informed patient preferences.	IIb	B
Beta-blockers should not be used routinely for the prevention of postoperative AF in patients undergoing non-cardiac surgery.	III	B

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Recommendations pertaining to sex-related differences in AF

Recommendations	Class	Level
It is recommended that women and men with AF are equally offered diagnostic assessment and therapies to prevent stroke and other AF-related complications.	I	A
Women with symptomatic paroxysmal or persistent AF should be offered timely access to rhythm control therapies, including AF catheter ablation, when appropriate for medical reasons.	IIa	B

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Table 22 Summary of quality indicators for the diagnosis and management of AF (1)

Domain: Patient assessment (at baseline and follow-up)
Main quality indicator: CHA_2DS_2-VASc cardioembolic risk assessment.
Main quality indicator: bleeding risk assessment using a validated method such as the HAS-BLED score.
Numerator: number of AF patients who have their respective score documented at the time of diagnosis and at every follow-up appointment. *Denominator*: number of AF patients.
Domain: Anticoagulation
Main quality indicator: inappropriate prescription of anticoagulation to patients with a CHA_2DS_2-VASc score of 0 for men and 1 for women.
Numerator: number of AF patients with CHA_2DS_2-VASc score of 0 for men and 1 for women, who are inappropriately prescribed anticoagulation. *Denominator*: number of AF patients with CHA_2DS_2-VASc score of 0 for men and 1 for women who do not have other indication for anticoagulation.

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Table 22 Summary of quality indicators for the diagnosis and management of AF (2)

Domain: Anticoagulation (continued)
Main quality indicator: proportion of patients with a CHA_2DS_2-VASc score of ≥1 for men and ≥2 for women who are prescribed anticoagulation.
Numerator: number of AF patients with CHA_2DS_2-VASc score of ≥1 for men and ≥2 for women who are prescribed anticoagulation. *Denominator*: number of AF patients with CHA_2DS_2-VASc score of ≥1 for men and ≥2 for women who are eligible for anticoagulation with no contraindication or refusal.
Domain: rate control
Main quality indicator: inappropriate prescription of AADs[a] to patients with permanent AF (i.e. where no attempt to restore sinus rhythm is planned).
Numerator: number of patients with permanent AF who are prescribed one or more AADs[a] for rhythm control. *Denominator*: number of patients with permanent AF.

[a]Flecainide, propafenone, amiodarone, dronedarone, sotalol and disopyramide.

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Table 22 Summary of quality indicators for the diagnosis and management of AF (3)

Domain: rhythm control
Main quality indicator: inappropriate prescription of class IC AADs to patients with structural heart disease.
Numerator: number of AF patients with structural heart disease who are inappropriately prescribed class IC AADs. *Denominator*: number of AF patients with structural heart disease.
Main quality indicator: proportion of patients with symptomatic paroxysmal or persistent AF who are offered AF catheter ablation after failure of/intolerance to one class I or class III AAD.
Numerator: number of patients with paroxysmal or persistent AF who are offered catheter ablation after the failure of, or intolerance to, at least one class I or class III AAD. *Denominator*: number of patients with paroxysmal or persistent AF with no contraindications (or refusal) to catheter ablation who remain symptomatic on, or intolerant to at least one class I or class III AAD.

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Table 22 Summary of quality indicators for the diagnosis and management of AF (4)

Domain: risk factor management
Main quality indicator: proportion of patients who have their modifiable risk factors identified.
Numerator: number of AF patients who have their modifiable risk factors (e.g. BP, obesity, OSA, alcohol excess, lack of exercise, poor glycaemic control and smoking) identified *Denominator*: number of AF patients.
Domain: outcomes
Main quality indicator: ischaemic stroke or TIA.
Main quality indicator: life-threatening or major bleeding events.[b]
Numerator: number of AF patients who have a documented ischaemic or bleeding event *Denominator*: number of AF patients or number of patients prescribed an OAC, respectively.

[b]Using the definitions of the International Society of Thrombosis and Haemostasis.

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Recommendations for quality measures in patients with AF

Recommendations	Class	Level
The introduction of tools to measure quality of care and identify opportunities for improved treatment quality and AF patient outcome should be considered by practitioners and institutions.	IIa	B

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Figure 24 Progression of atrial high-rate episode burden (left panel) and stroke rates according to AHRE daily burden and CHA_2DS_2-VASc score (right panel)

Six-month incidence of transition to higher AHRE burden[a]
(*n* = 6580, pooled from three prospective studies)

	Baseline burden			
6-month progression	5 min to <1 h	1 h to <6 h	6 h to <12 h	12 h to <23 h
Transition to ≥1 h	33.5%			
Transition to ≥6 h	15.3%	42.2%		
Transition to ≥12 h	8.9%	27.5%	55.8%	
Transition to ≥23 h	5.1%	16.0%	40.6%	63.1%

Stroke rates[b] per AHRE burden and CHA_2DS_2-VASc category
(*n* = 21 768 device patients not taking OAC)

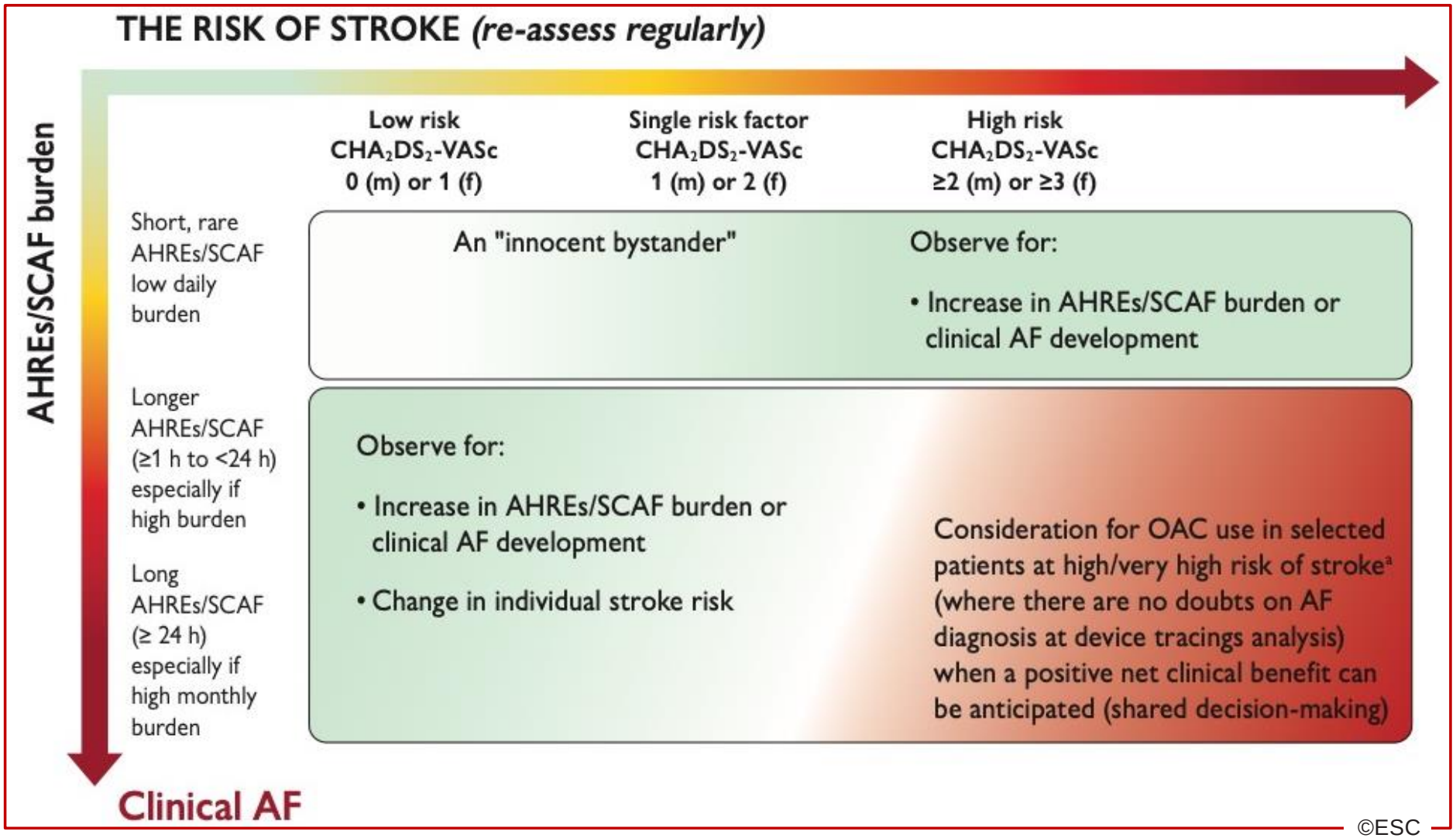
	Baseline maximum daily burden		
CHA_2DS_2-VASc score	No AF	AF 6 min–23.5 h	AF >23.5 h
0	0.33%	0.52%	0.86%
1	0.62%	0.32%	0.50%
2	0.70%	0.62%	1.52%
3-4	0.83%	1.28%	1.77%
≥5	1.79%	2.21%	1.68%

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[a]The higher the burden at diagnosis, the greater the incidence of progression in the next 6 months and thereafter. [b]Stroke rates above the threshold for OAC are shown in red.

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Figure 25 Proposed management of AHRE/subclinical AF



[a]Highly selected patients (e.g. with previous stroke and/or age ≥75 years, or ≥3 CHA_2DS_2-VASc risk factors, and additional non-CHA_2DS_2-VASc stroke factors such as CKD, elevated blood biomarkers, spontaneous echo contrast in dilated LA, etc); selected patients (e.g. with previous stroke and/or age ≥75 years, or ≥3 CHA_2DS_2-VASc risk factors , etc).

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Recommendations for management of patients with AHRE

Recommendations	Class	Level
In patients with AHRE/subclinical AF detected by CIED or insertable cardiac monitor, it is recommended to conduct: • Complete cardiovascular evaluation with ECG recording, clinical risk factors/comorbidity evaluation, and thromboembolic risk assessment using the CHA_2DS_2-VASc score. • Continued patient follow-up and monitoring (preferably with the support of remote monitoring) to detect progression to clinical AF, monitor the AHRE/subclinical AF burden (especially transition to ≥24 hours), and detect changes in underlying clinical conditions.	I	B

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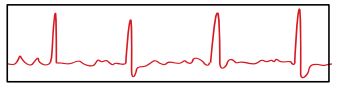
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Central Illustration Management of AF (1)

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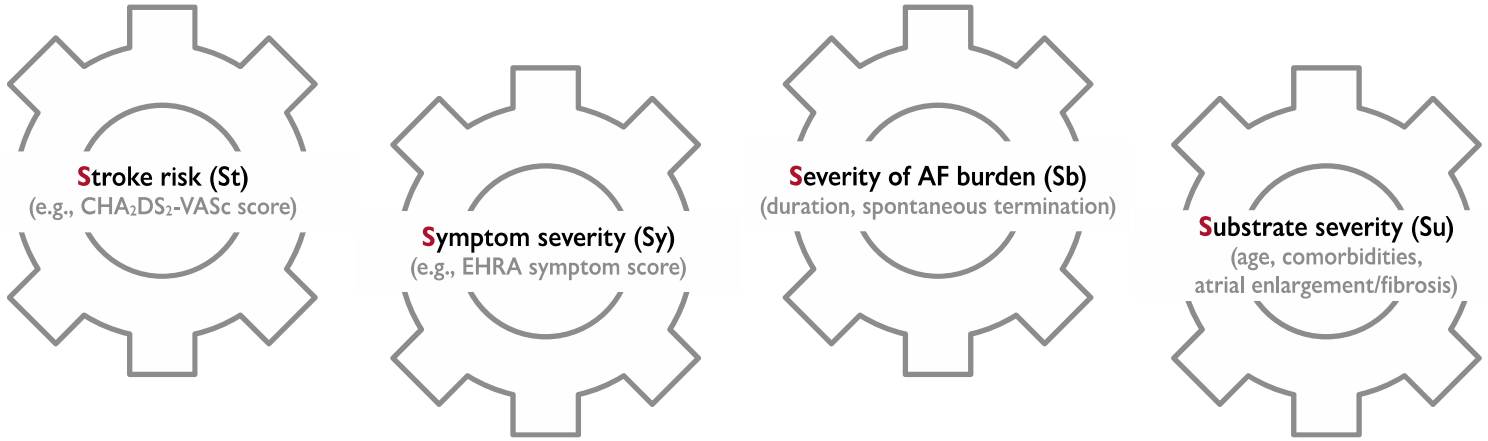
CC To ABC

Confirm AF



A 12-lead ECG or a rhythm strip showing AF pattern for ≥30 s

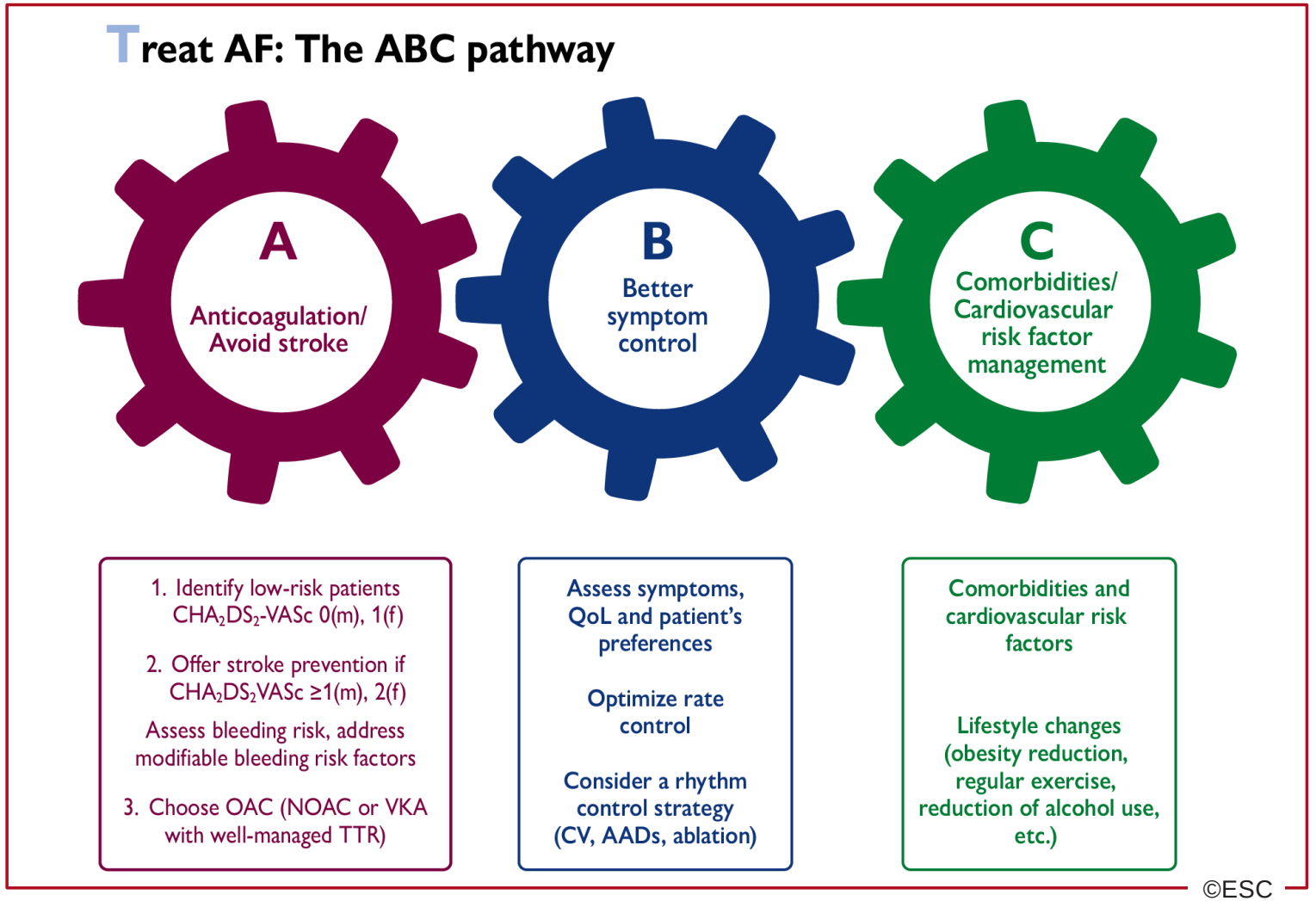
Charactirize AF (the 4S-AF scheme)



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Central Illustration Management of AF (2)



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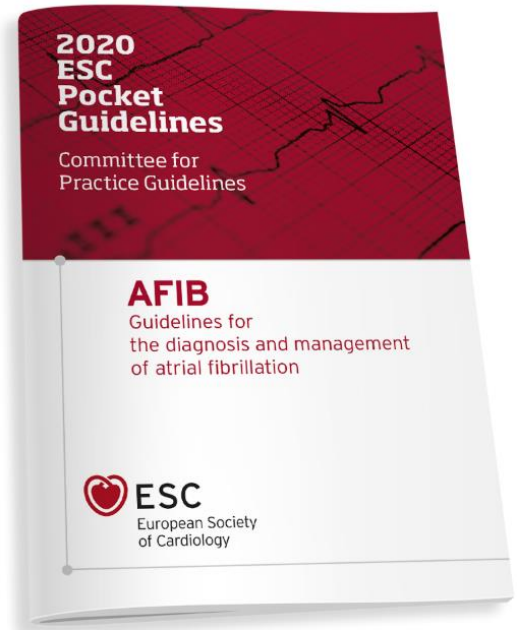


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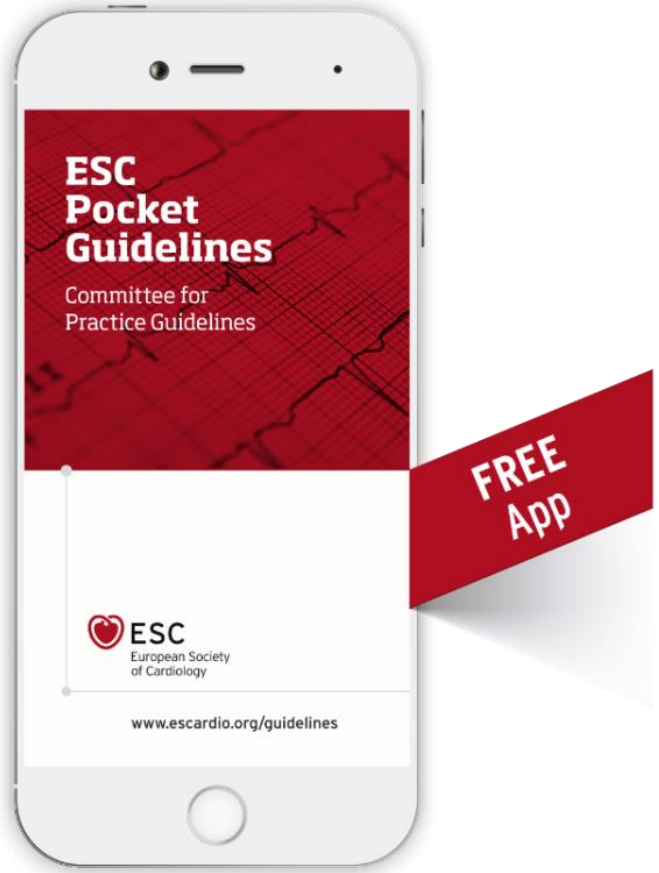
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